

Step-7 (Track A) — Horizon x Cov-Duration Bake-off Master Dashboard (Sigma 30s / 45s / 60s, MASKED)

2022-2025 · cov-dur x horizon(nextwin..1h) x metric(LE/AI) x 15 learners · fixed Step-6 predictor [rv, s_slow, s_fast, v]; realized-VOL target aggregated to horizon · AI headline / LE diagnostic · next-window COINCIDENT, no next-day forecast (deferred Track B) · cells finished: 20 · durations present: 30s

What this run is. A **cov-duration x prediction-horizon** bake-off. No new covariance is built: the 30/45/60 s **masked Sigma state** (Steps 3-6) is unchanged; Step 7 moves ONE axis - the target's forward SPAN H (nextwin, 2m, 5m, 10m, 15m, 30m, 1h). For every (cov-dur x horizon x metric) cell we fit the COMPLETE 15-learner panel (GridSearch + purged walk-forward) with the **fixed Step-6 predictor** [rv, s_slow, s_fast, v] and a realized-VOL target aggregated to H. **LE** features = Log-Euclidean Step-6 dial; **AI** features = affine-invariant dial (PC1 of the AI whitened tangent) + AI geodesic velocity **AI** **LE** . **Every number below is labelled with its cov-duration AND its horizon.**

The claim being tested. Holding the covariance state fixed: (1) at which HORIZON does the dial stop being coincident and start adding OOS value over persistence (the **dr2(H)** curve); (2) does a FLEXIBLE learner beat linear at that horizon (capacity); (3) is the answer METRIC-ROBUST (AI vs LE)? **next-window is COINCIDENT** (persistence wins); longer horizons test whether the dial becomes LEADING. There is NO next-day forecast here - that is the deferred Track B.

30s
cov durations

nextwin . 30s . 45s . 60s . 2m . 5m . 10m . 15m . 30m . 1h
horizons

LE / AI
metrics

15
learners / cell

20 / 20
cells finished

0
cells N/A (skipped)

30s
durations present

Coverage grid (cov-duration x horizon). which cells the detached run has finished (LE+AI green / one metric amber), which are physically SKIPPED (N/A = horizon finer than the cov window), and which are still pending (—).

Sigma \ horizon	nextwin	30s	45s	60s	2m	5m	10m	15m	30m	1h
30s	LE+AI	LE+AI	LE+AI	LE+AI	LE+AI	LE+AI	LE+AI	LE+AI	LE+AI	LE+AI

Peak lead-vs-coincident edge (per cov-dur x metric). the horizon where the best learner's dr2 over persistence is largest, and whether its 95% CI clears 0.

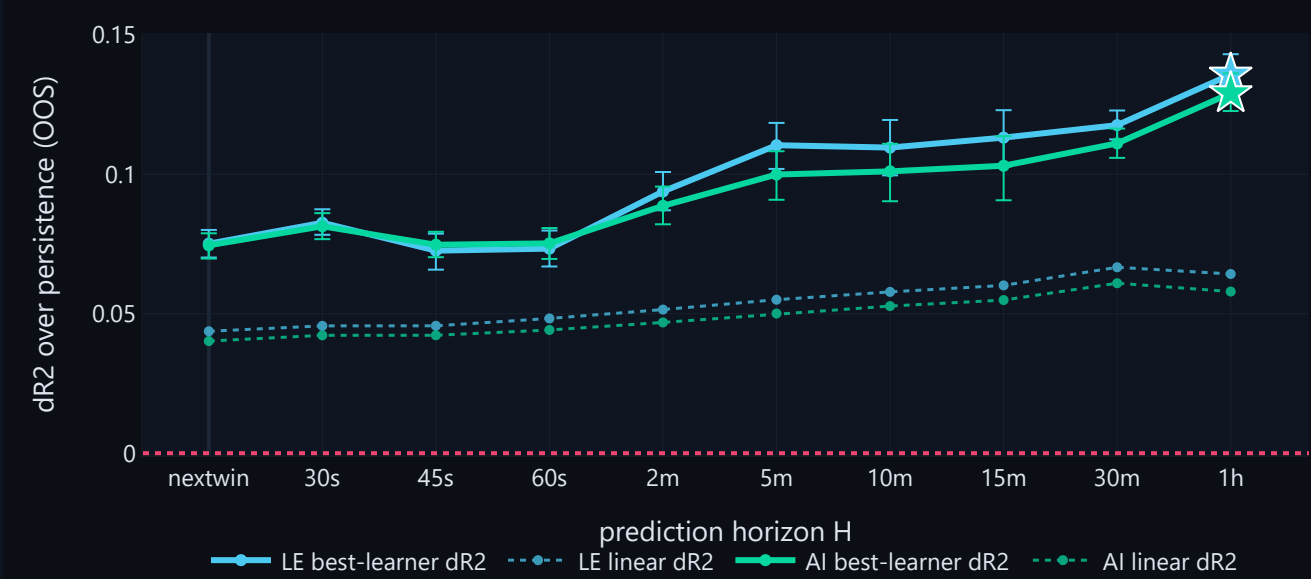
Sigma	metric	horizons done	peak dr2 horizon	peak dr2	sig (CI>0)
30s	LE	10 / 10	1h	0.136	True
30s	AI	10 / 10	1h	0.129	True

THE headline of Track A - lead-vs-coincident. For each cov-duration, plot the dial's OOS edge over persistence **dR2(horizon)** as the target span H grows (x = nextwin, 2m, 5m, 10m, 15m, 30m, 1h). Solid = the BEST learner's dR2; dotted = the linear learner's dR2. LE (blue) vs AI (green) **AI** **LE** . **The question:** at next-window the dial is COINCIDENT (persistence wins, dR2 small); does dR2 GROW with horizon (dial becomes LEADING)? The marked star = the horizon that PEAKS dR2 (the characteristic horizon).

dR2(horizon) per cov-duration — native 30s / 45s / 60s Sigma; LE vs AI

Sigma=30s dR2(horizon) lead-vs-coincident

dR2(horizon) -- lead-vs-coincident [Sigma=30s]



In plain terms How much extra vol-forecast accuracy the covariance dial adds over pure persistence, at each forward span.

Math / formula $dR2(H) = R2_OOS(\text{best learner}) - R2_OOS(\text{persistence=rv-only})$; block-bootstrap 95% CI over days.

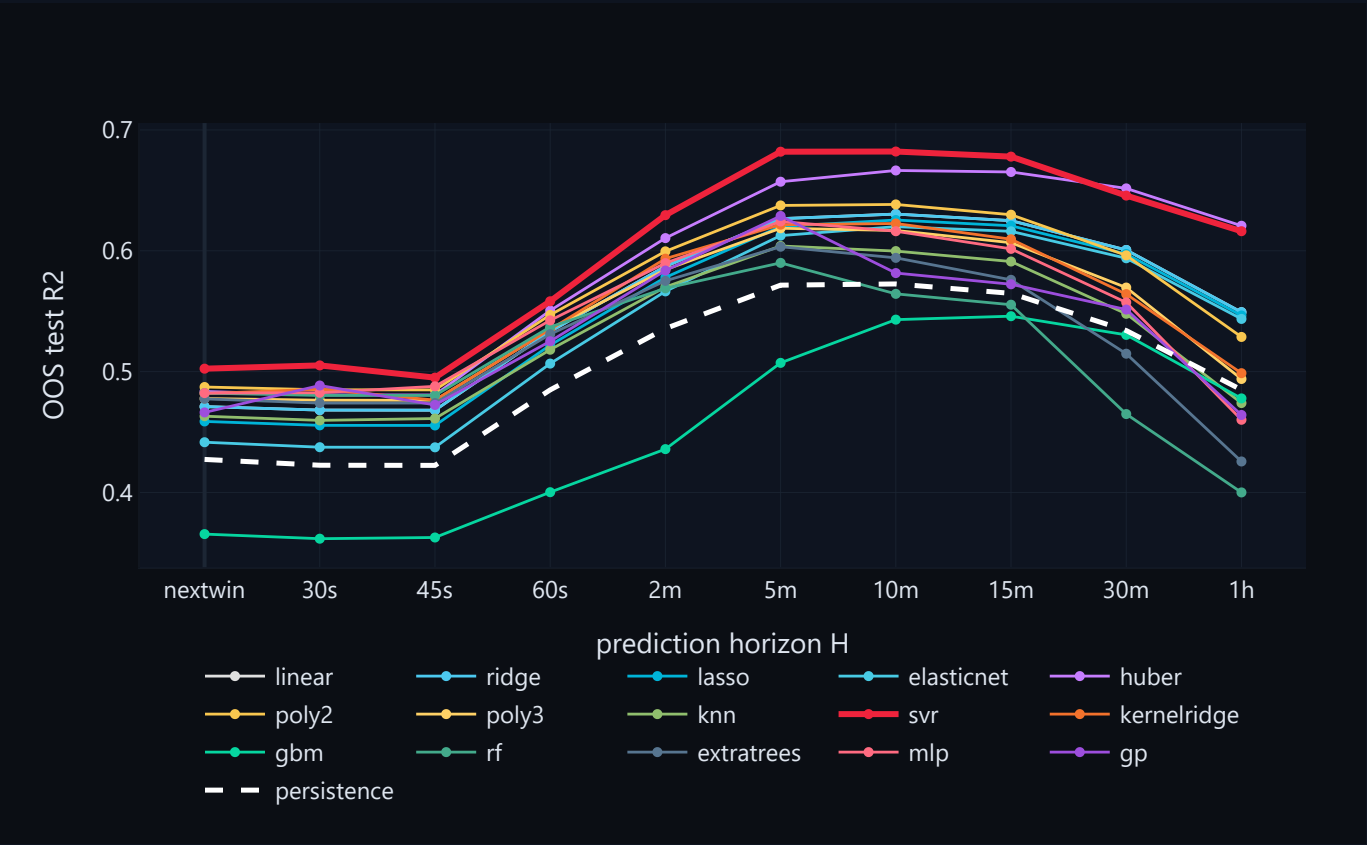
How to read it flat/low near nextwin = coincident; rising with H = the dial becomes leading; star marks the peak H.

Insight LE: peak dR2 **0.136** @ 1h (nextwin 0.075) [sig] | AI: peak dR2 **0.129** @ 1h (nextwin 0.075) [sig]

Performance-comparison plots. The same numbers as the tables, drawn for the eye. Every chart is labelled with its cov-duration / horizon / metric; horizons are always in canonical order (nextwin, 30s, 45s, 60s, 2m, 5m, 10m, 15m, 30m, 1h); skipped cells (horizon finer than the cov window) are greyed/omitted. Colors = learners (svr = the usual winner, highlighted).

1 · OOS test R2 vs horizon — 15 learners + persistence — one chart per cov-duration x metric

Sigma=30s OOS R2 vs horizon [Sigma=30s | metric=LE] LE



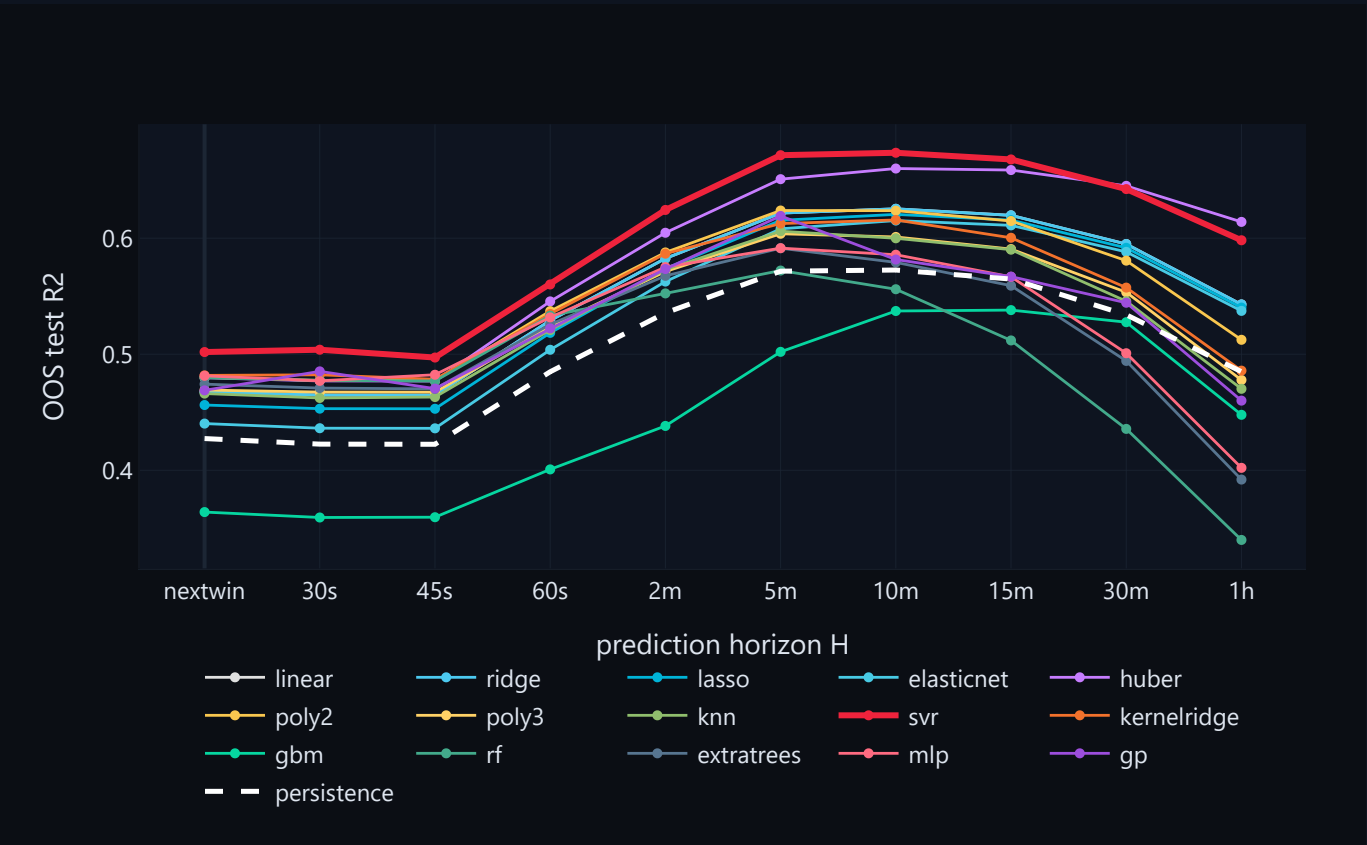
In plain terms How each of the 15 learners tracks vol as the forecast span grows, vs the persistence floor.

Math / formula $y = R2_OOS(model)$ per horizon; dashed white = persistence (rv-only) baseline.

How to read it lines above the dashed floor beat persistence; watch the interior peak (~10m).

Insight cov 30s, metric LE; missing points = unfinished/skipped horizons.

Sigma=30s OOS R2 vs horizon [Sigma=30s | metric=AI] AI



In plain terms How each of the 15 learners tracks vol as the forecast span grows, vs the persistence floor.

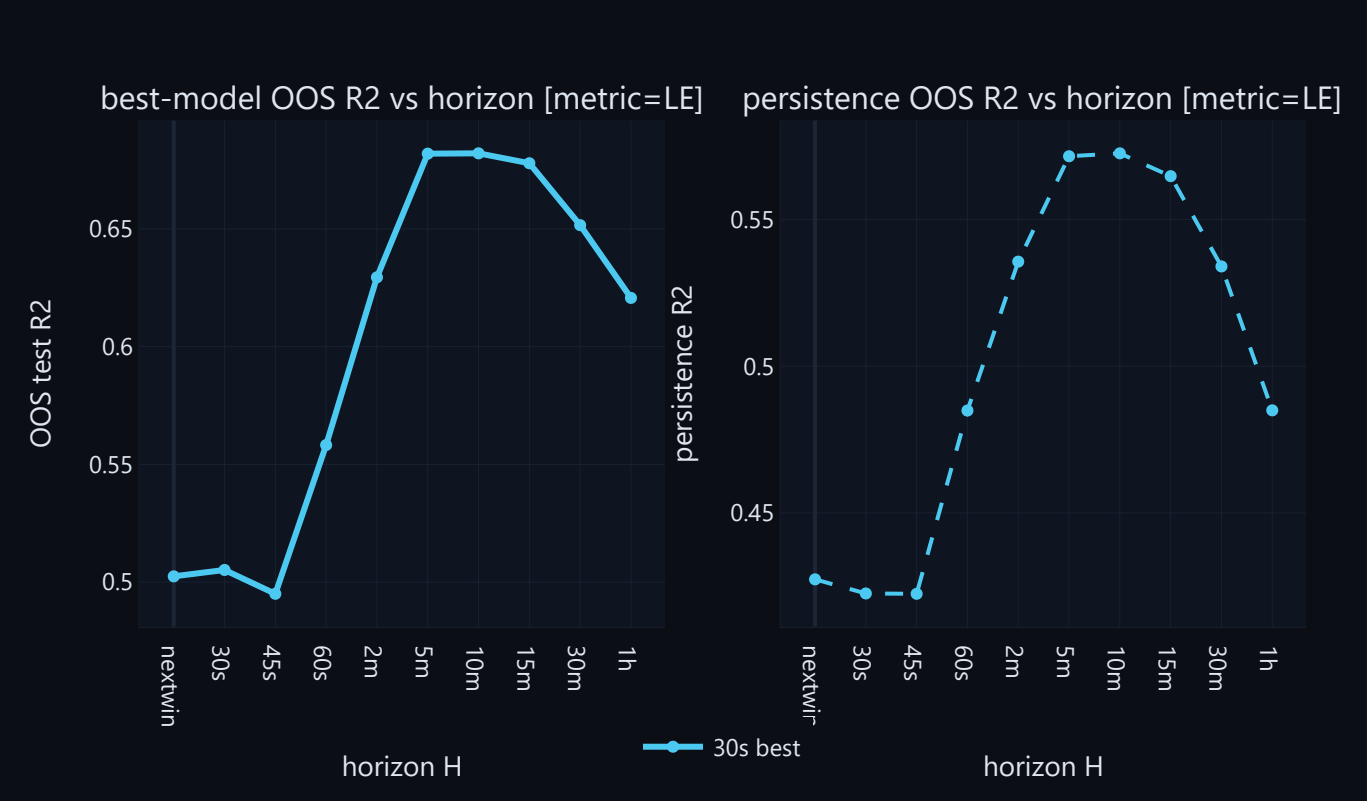
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How to read it lines above the dashed floor beat persistence; watch the interior peak (~10m).

Insight cov 30s, metric AI; missing points = unfinished/skipped horizons.

3 · Cross-duration overlay — 30s vs 45s vs 60s — best-model R2 (left) and persistence R2 (right) vs horizon

Sigma=30s Cross-duration: best-model & persistence R2 vs horizon [metric=LE] LE



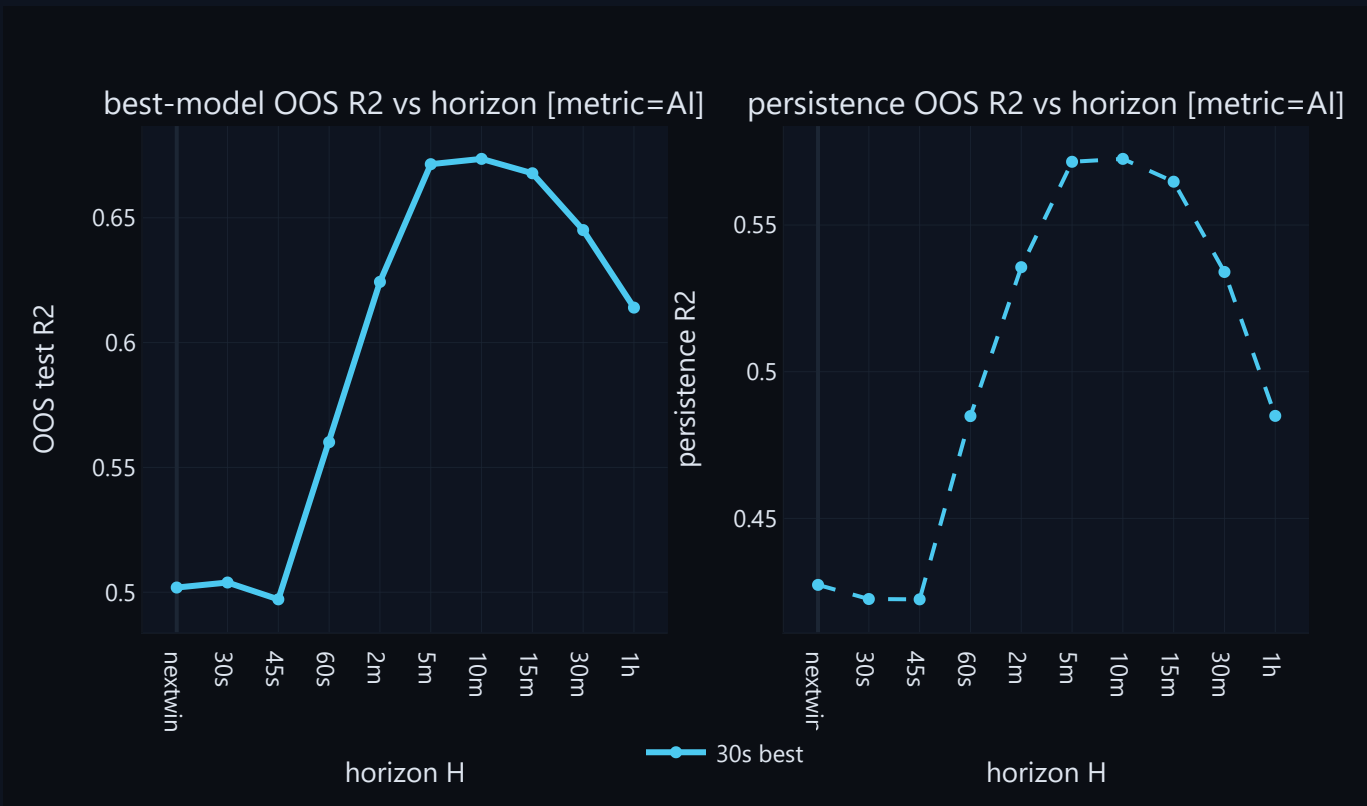
In plain terms Does a coarser covariance window (60s) give a higher absolute R2 than a finer one (30s)?

Math / formula left: best-learner R2 per horizon for 30/45/60s; right: the persistence floor per horizon.

How to read it
Insight

curves ordered 60s>45s>30s = coarser Sigma is more predictable (both model and floor rise).
metric LE; both panels share the horizon axis.

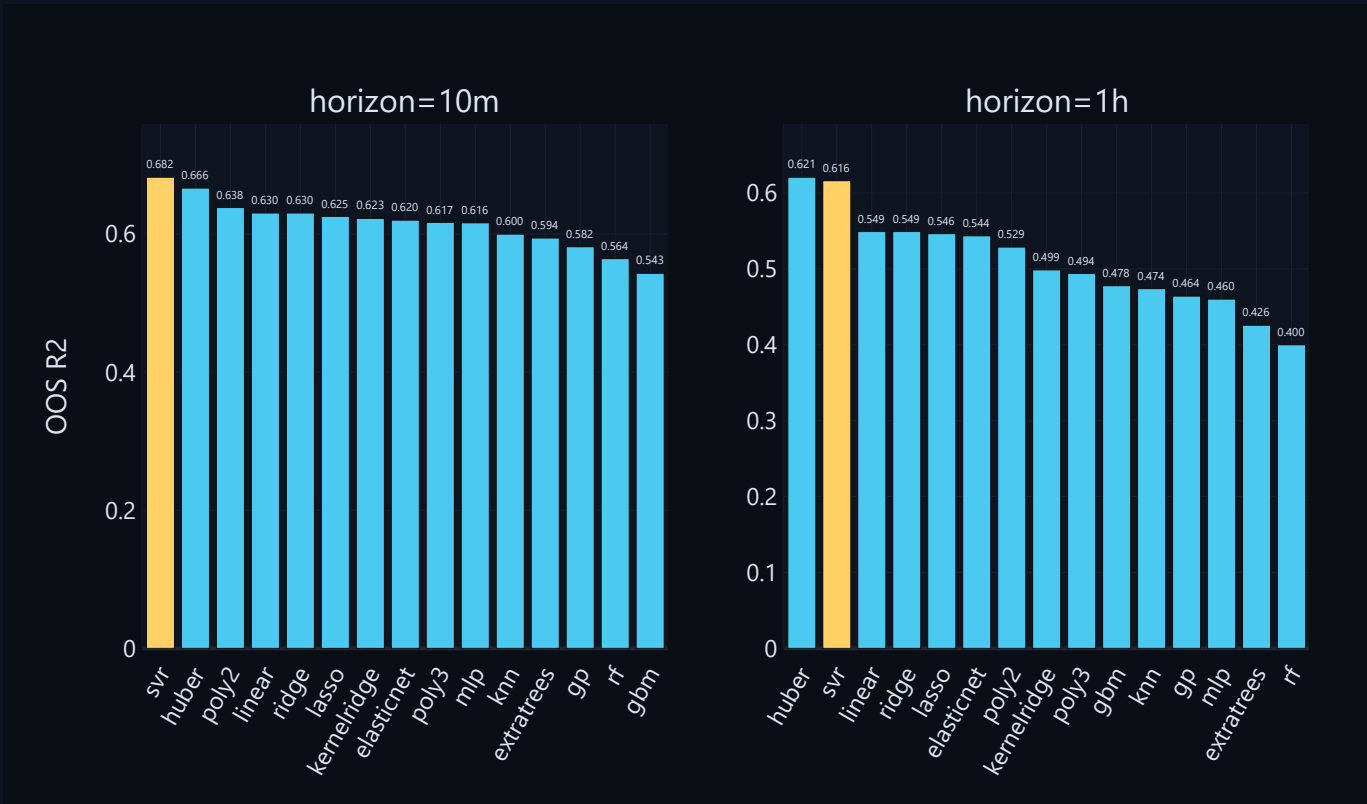
Sigma=30s Cross-duration: best-model & persistence R2 vs horizon [metric=AI] AI



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How to read it curves ordered 60s>45s>30s = coarser Sigma is more predictable (both model and floor rise).
Insight metric AI; both panels share the horizon axis.

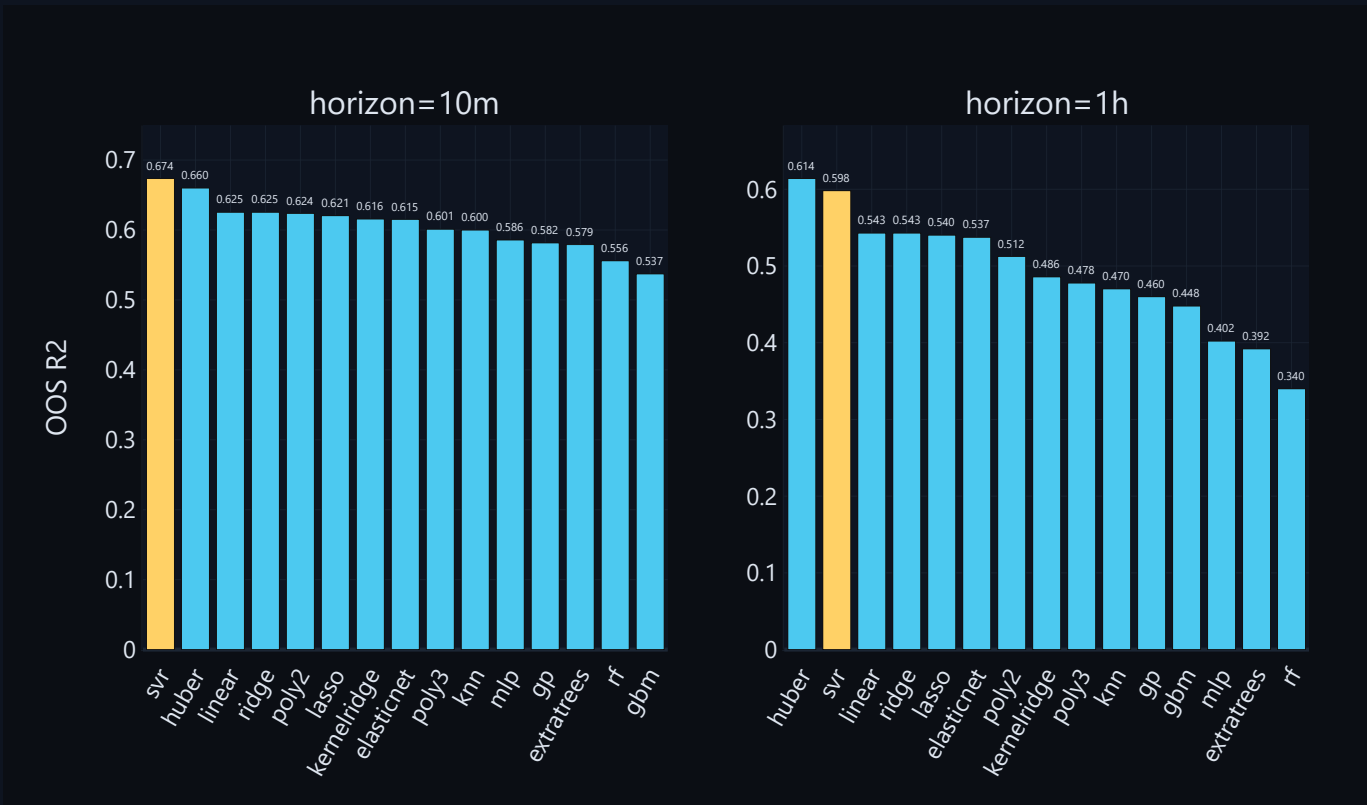
4 · Model ranking (OOS R2, sorted) — at 10m and 1h — per cov-duration x metric; winner highlighted

Sigma=30s Learner ranking [Sigma=30s | metric=LE] LE



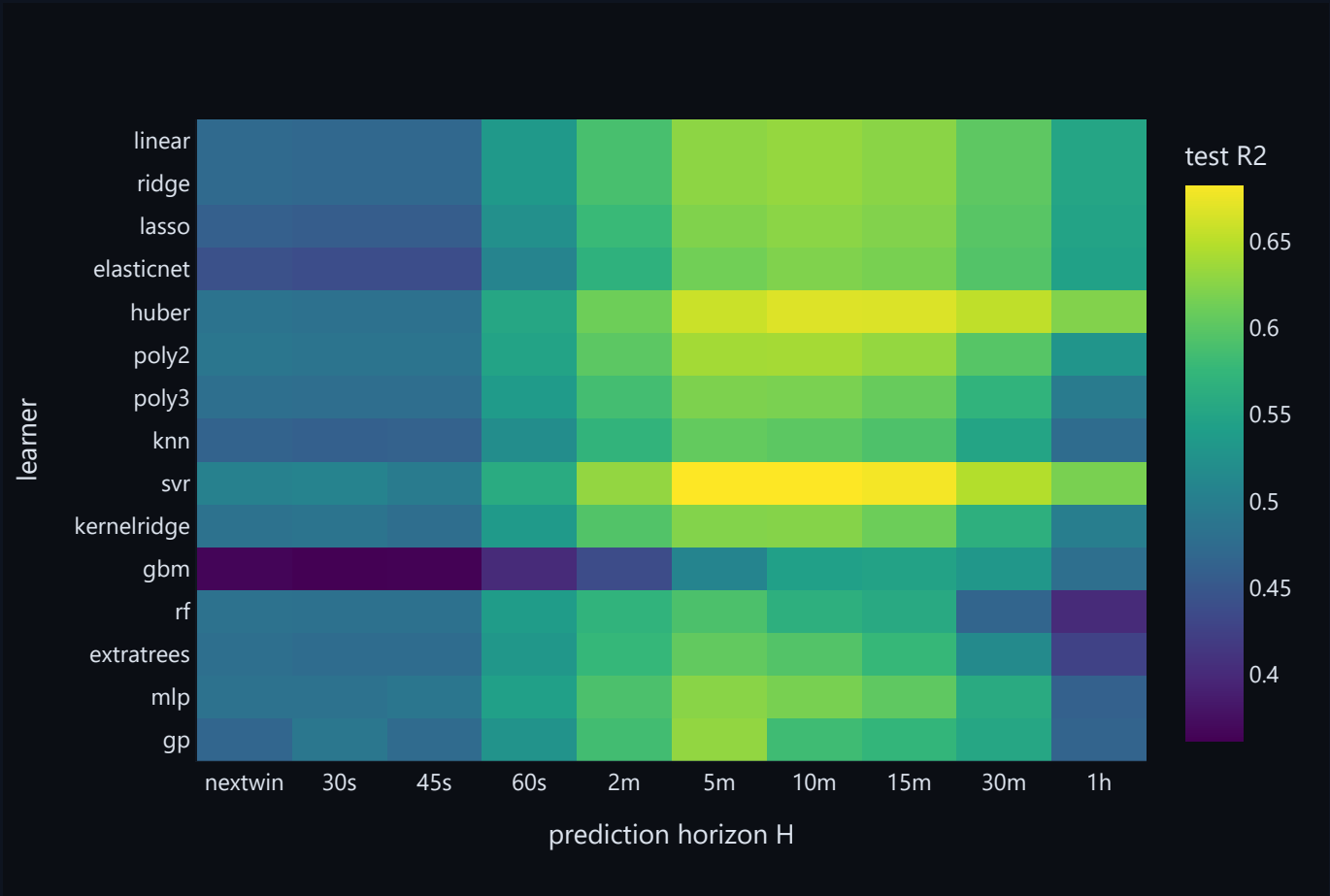
In plain terms Which learner wins at the interesting horizons, and by how much.
Math / formula bars = R2_OOS per learner, sorted descending; amber = svr (the usual winner).
How to read it a tall gap from #1 to linear = capacity helps; a flat top = many models tie.
Insight cov 30s, metric LE, horizons 10m, 1h.

Sigma=30s Learner ranking [Sigma=30s | metric=AI] AI



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How to read it a tall gap from #1 to linear = capacity helps; a flat top = many models tie.
Insight cov 30s, metric AI, horizons 10m, 1h.

Sigma=30s OOS R2 heatmap [Sigma=30s | metric=LE] LE



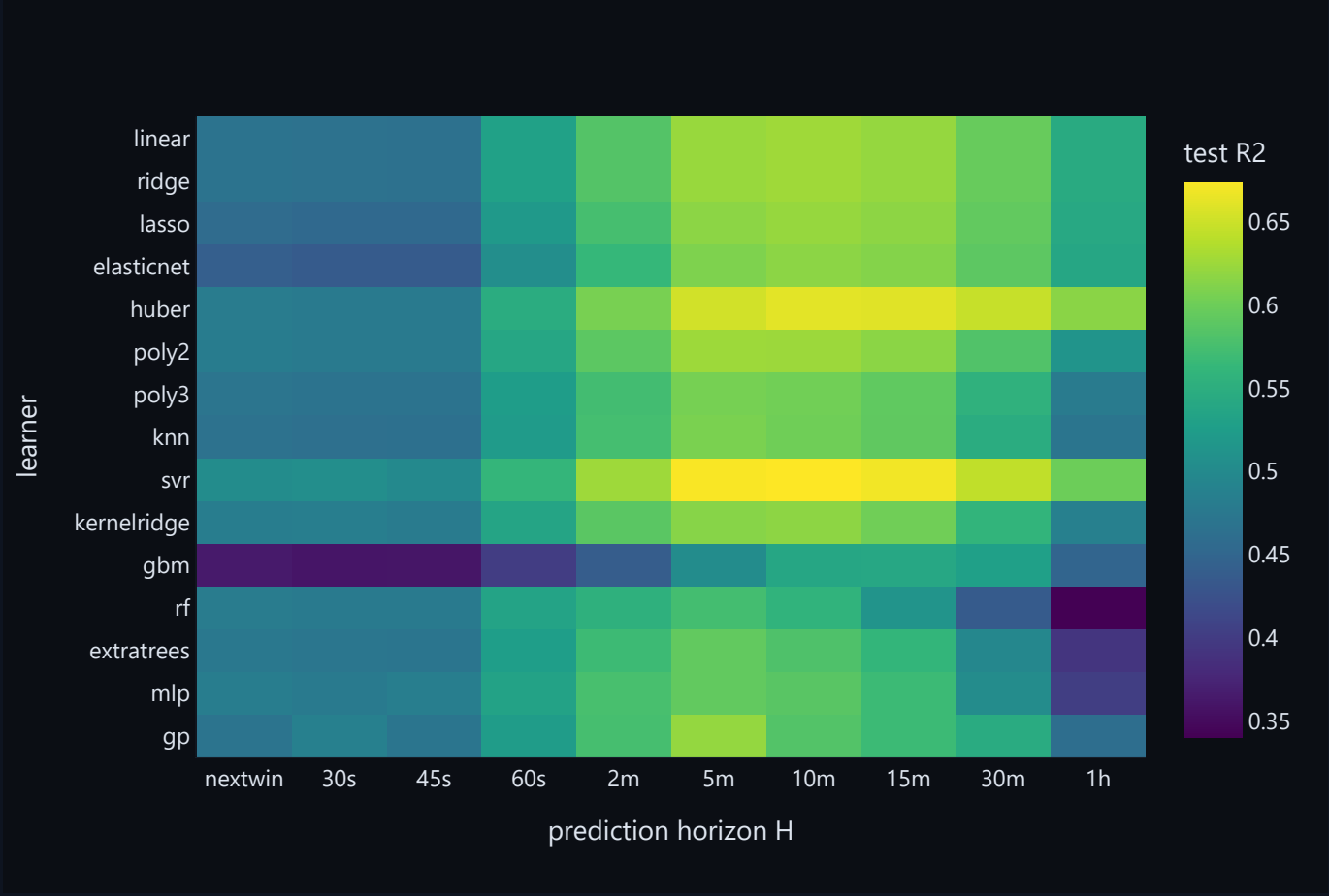
In plain terms The whole learner x horizon accuracy grid at a glance.

Math / formula cell colour = $R2_OOS(learner, horizon)$; grey = skipped (horizon < cov window).

How to read it bright bands = strong learner/horizon combos; a bright column = an easy horizon.

Insight cov 30s, metric LE; 15 learners x 10 finished horizons.

Sigma=30s OOS R2 heatmap [Sigma=30s | metric=AI] AI



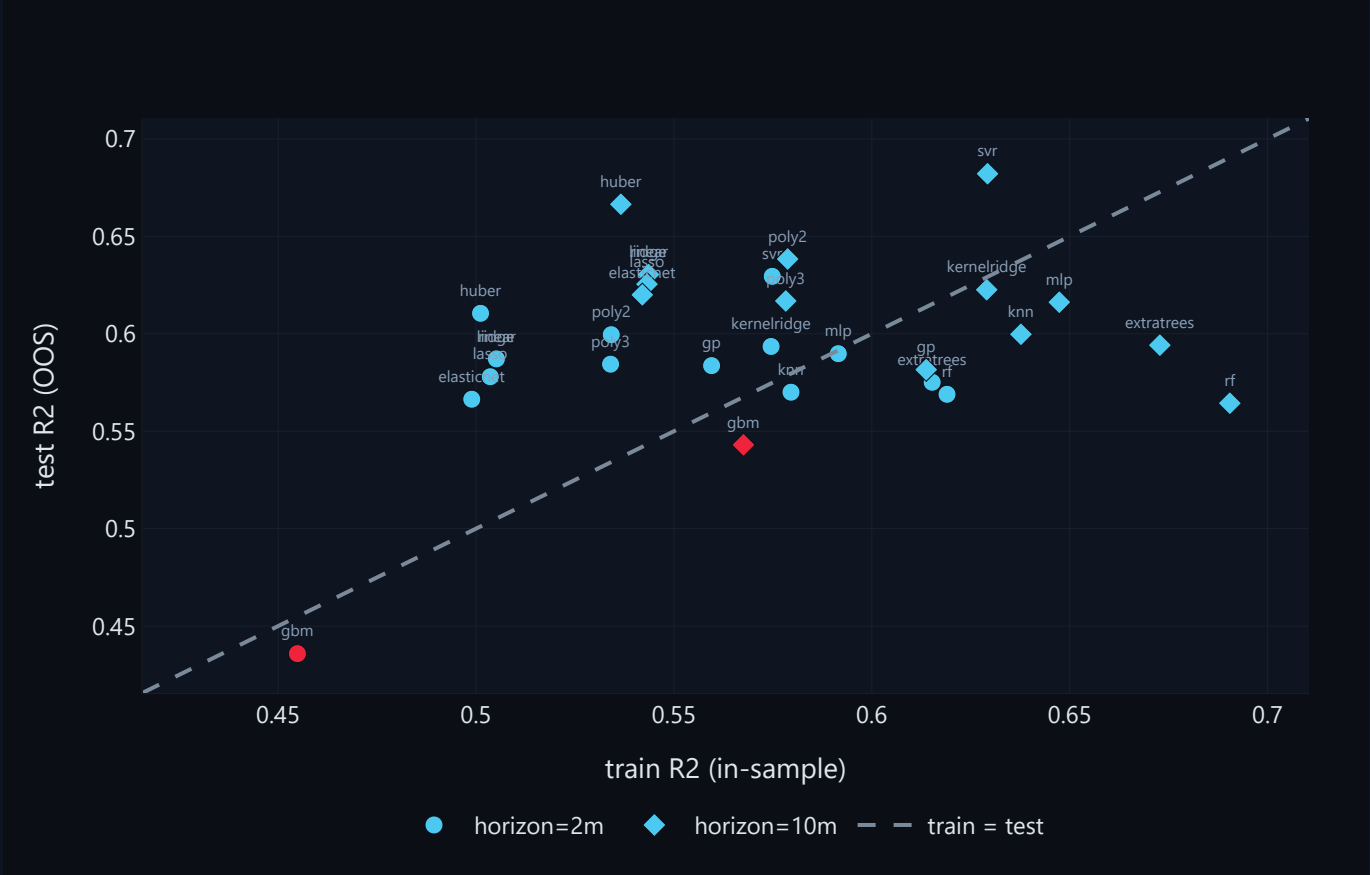
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Sigma=30s Overfit: train vs test R2 [Sigma=30s | metric=LE] LE



In plain terms

Math / formula

How to read it

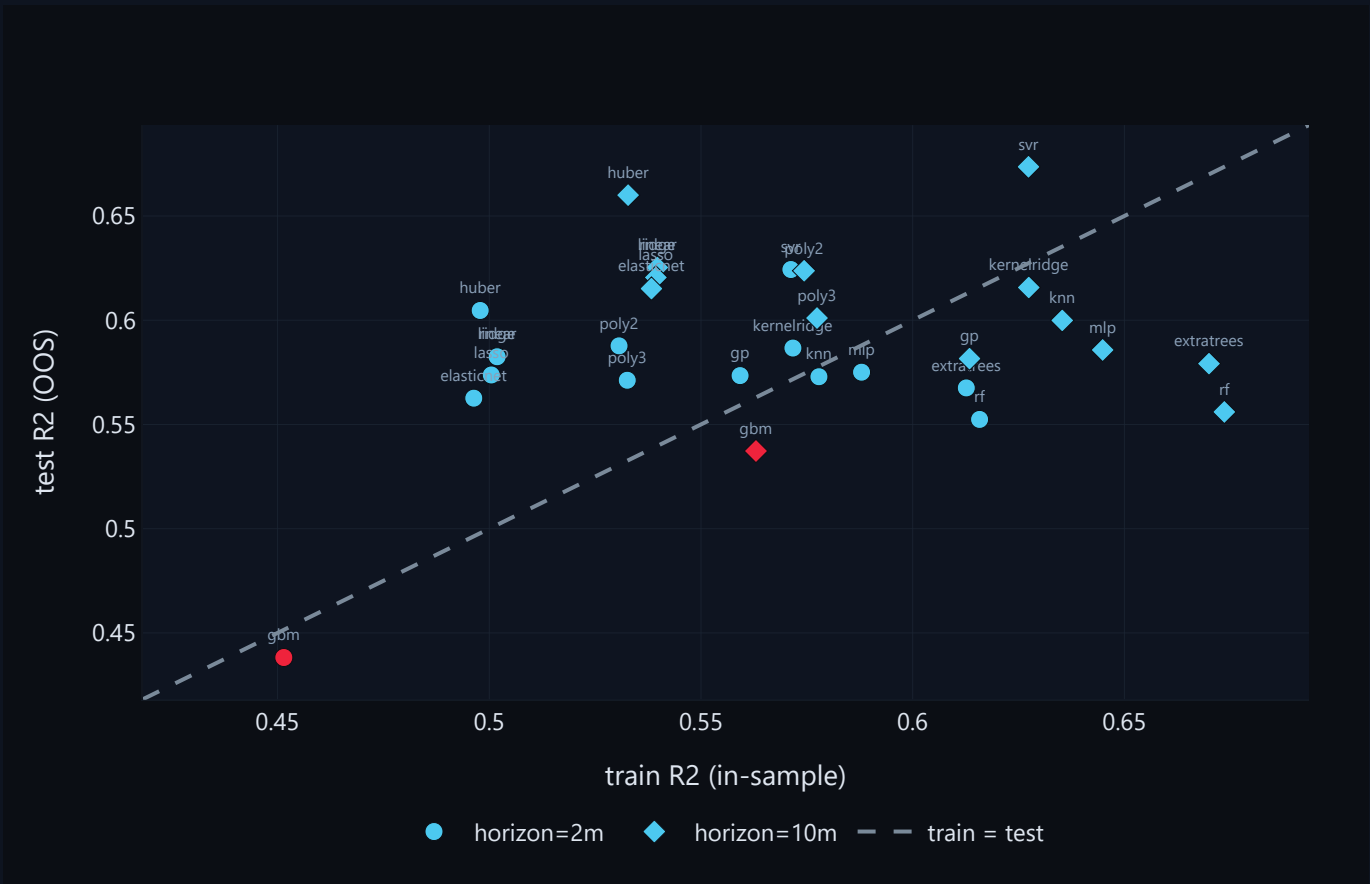
Insight

How much each learner's in-sample fit exceeds its out-of-sample accuracy (the gap = overfit).
x = train R2, y = test R2; dashed = no gap; distance BELOW the line = overfit.
points far under the diagonal (watch gbm, red) overfit; on/near it generalise.
cov 30s, metric LE, horizons 2m, 10m.

Sigma=30s

Overfit: train vs test R2 [Sigma=30s | metric=AI]

AI



In plain terms

Math / formula

How to read it

Insight

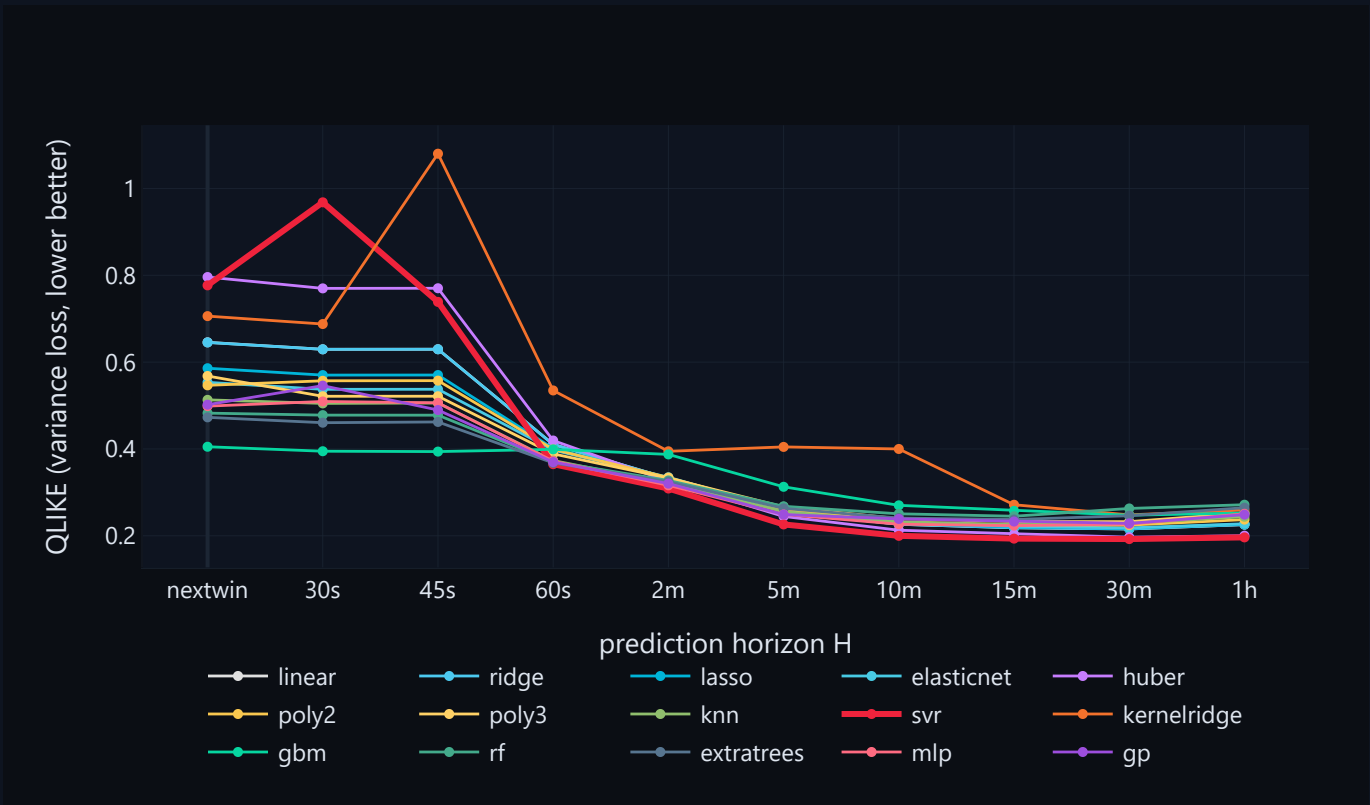
How much each learner's in-sample fit exceeds its out-of-sample accuracy (the gap = overfit).
x = train R2, y = test R2; dashed = no gap; distance BELOW the line = overfit.
points far under the diagonal (watch gbm, red) overfit; on/near it generalise.
cov 30s, metric AI, horizons 2m, 10m.

8 · QLIKE vs horizon — 15 learners (lower is better) — per cov-duration x metric

Sigma=30s

QLIKE vs horizon [Sigma=30s | metric=LE]

LE



In plain terms

Math / formula

How to read it

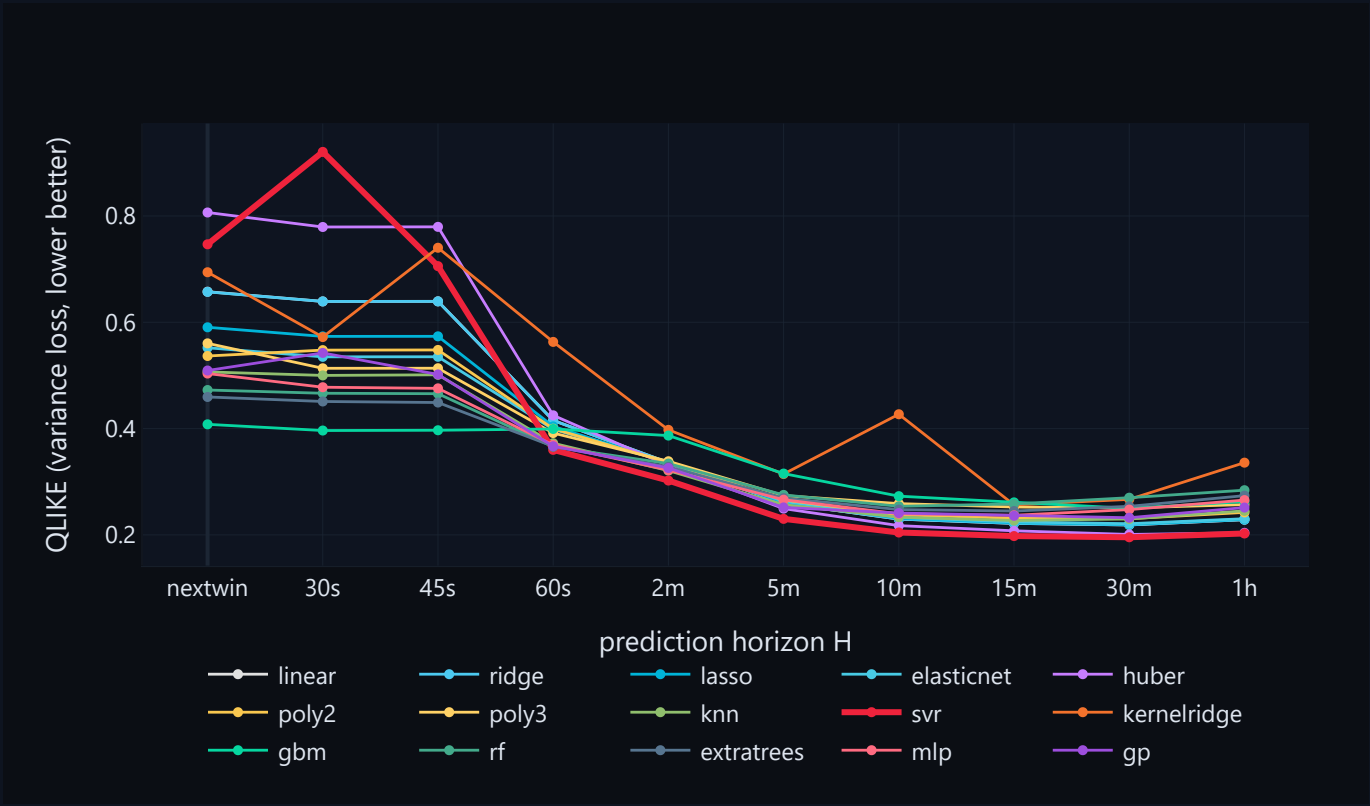
Insight

Does the accuracy story survive the stricter QLIKE variance loss (not just R2)?
 $y = \text{QLIKE_OOS}(\text{model}) \text{ per horizon} = \text{mean}(r - \ln r - 1)$, r = variance ratio; lower = better.
lines that stay low across horizons are robust; spikes flag loss-sensitive learners.
cov 30s, metric LE.

Sigma=30s

QLIKE vs horizon [Sigma=30s | metric=AI]

AI



In plain terms

Does the accuracy story survive the stricter QLIKE variance loss (not just R2)?

Math / formula
How to read it
Insight

y = QLIKE_OOS(model) per horizon = mean(r - ln r - 1), r = variance ratio; lower = better.
lines that stay low across horizons are robust; spikes flag loss-sensitive learners.
cov 30s, metric AI.

The full matrix. For EVERY (cov-dur x horizon x metric) cell, all 15 learners x {train R2, test R2, gap, dR2 [CI95], sig, RMSE, MAE, QLIKE, bias}. Grouped by cov-duration; within each, a labelled block per horizon with LE and AI side by side. Every table title carries **Sigma=<dur>s | horizon=<H> | metric=<LE/AI>**. The star row = best-by-test-R2. RMSE/MAE/bias in vol units; QLIKE on variance.

Sigma=30s cov-duration 30s — 10 horizon(s) finished

horizon=nextwin Sigma=30s · horizon=nextwin (H=30s) — metrics: LE + AI

LE Sigma=30s | horizon=nextwin | metric=LE

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
linear	0.404	0.471	-0.067	0.044 [0.040, 0.047]	yes	2.31e-04	1.85e-04	5.33e-08	0.646	5.32e-05	2.75e-04	2.12e-04	7.54e-08	0.901	1.41e-16
ridge	0.404	0.471	-0.067	0.044 [0.040, 0.047]	yes	2.31e-04	1.85e-04	5.33e-08	0.645	5.33e-05	2.75e-04	2.12e-04	7.54e-08	0.901	-5.96e-17
lasso	0.401	0.459	-0.058	0.031 [0.029, 0.034]	yes	2.34e-04	1.87e-04	5.45e-08	0.586	5.74e-05	2.75e-04	2.12e-04	7.59e-08	0.816	-5.63e-21
elasticnet	0.390	0.442	-0.052	0.014 [0.012, 0.016]	yes	2.37e-04	1.89e-04	5.63e-08	0.553	6.13e-05	2.78e-04	2.14e-04	7.72e-08	0.765	-1.90e-20
huber	0.402	0.484	-0.082	0.056 [0.054, 0.059]	yes	2.28e-04	1.82e-04	5.20e-08	0.796	3.58e-05	2.75e-04	2.11e-04	7.57e-08	1.110	-1.69e-05
poly2	0.425	0.487	-0.062	0.060 [0.055, 0.065]	yes	2.27e-04	1.82e-04	5.17e-08	0.547	5.83e-05	2.70e-04	2.08e-04	7.27e-08	0.766	4.62e-15
poly3	0.423	0.478	-0.054	0.050 [0.047, 0.054]	yes	2.29e-04	1.84e-04	5.26e-08	0.568	6.19e-05	2.70e-04	2.09e-04	7.30e-08	0.816	1.45e-13
knn	0.455	0.463	-0.008	0.036 [0.029, 0.042]	yes	2.33e-04	1.86e-04	5.41e-08	0.513	6.78e-05	2.63e-04	2.02e-04	6.90e-08	0.715	-1.32e-06
svr ★	0.453	0.502	-0.050	0.075 [0.070, 0.080]	yes	2.24e-04	1.79e-04	5.01e-08	0.777	5.10e-05	2.63e-04	2.02e-04	6.93e-08	1.032	-1.58e-05
kernelridge	0.452	0.482	-0.029	0.054 [0.045, 0.062]	yes	2.29e-04	1.83e-04	5.22e-08	0.706	6.52e-05	2.63e-04	2.02e-04	6.93e-08	0.986	-1.25e-06
gbm	0.360	0.366	-0.005	-0.062 [-0.070, -0.052]	no	2.53e-04	2.03e-04	6.39e-08	0.405	9.48e-05	2.85e-04	2.20e-04	8.10e-08	0.532	1.26e-08
rf	0.467	0.482	-0.016	0.055 [0.048, 0.062]	yes	2.28e-04	1.84e-04	5.22e-08	0.483	6.78e-05	2.60e-04	2.01e-04	6.75e-08	0.656	-2.98e-08
extratrees	0.492	0.477	0.015	0.050 [0.043, 0.056]	yes	2.29e-04	1.85e-04	5.27e-08	0.473	7.10e-05	2.53e-04	1.95e-04	6.43e-08	0.596	-8.80e-22
mlp	0.466	0.482	-0.017	0.055 [0.047, 0.062]	yes	2.28e-04	1.84e-04	5.22e-08	0.499	7.02e-05	2.60e-04	2.00e-04	6.76e-08	0.671	1.92e-06
gp	0.444	0.466	-0.023	0.039 [0.032, 0.045]	yes	2.32e-04	1.86e-04	5.38e-08	0.502	7.33e-05	2.65e-04	2.04e-04	7.04e-08	0.718	-5.28e-08

best-by-test-R2 = svr (0.502) · linear 0.471 · persistence 0.427 · H=30s · 2774.4s. te.* = OOS test losses; tr.* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

AI Sigma=30s | horizon=nextwin | metric=AI

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
linear	0.402	0.468	-0.066	0.040 [0.037, 0.044]	yes	2.32e-04	1.85e-04	5.37e-08	0.658	5.28e-05	2.75e-04	2.13e-04	7.57e-08	0.904	-4.56e-16
ridge	0.402	0.468	-0.066	0.040 [0.037, 0.044]	yes	2.32e-04	1.85e-04	5.37e-08	0.657	5.29e-05	2.75e-04	2.13e-04	7.57e-08	0.904	-6.13e-17
lasso	0.398	0.456	-0.058	0.029 [0.026, 0.032]	yes	2.34e-04	1.87e-04	5.48e-08	0.591	5.76e-05	2.76e-04	2.13e-04	7.61e-08	0.818	7.04e-22
elasticnet	0.389	0.440	-0.051	0.013 [0.011, 0.015]	yes	2.37e-04	1.90e-04	5.64e-08	0.552	6.23e-05	2.78e-04	2.14e-04	7.73e-08	0.763	-1.62e-20
huber	0.399	0.480	-0.081	0.052 [0.050, 0.055]	yes	2.29e-04	1.83e-04	5.24e-08	0.807	3.56e-05	2.76e-04	2.12e-04	7.61e-08	1.117	-1.69e-05
poly2	0.422	0.480	-0.058	0.053 [0.048, 0.058]	yes	2.29e-04	1.84e-04	5.24e-08	0.537	5.98e-05	2.70e-04	2.09e-04	7.31e-08	0.757	3.14e-15
poly3	0.421	0.469	-0.048	0.042 [0.037, 0.046]	yes	2.31e-04	1.86e-04	5.35e-08	0.560	6.50e-05	2.71e-04	2.09e-04	7.32e-08	0.792	-1.27e-13
knn	0.453	0.466	-0.013	0.039 [0.033, 0.044]	yes	2.32e-04	1.86e-04	5.38e-08	0.507	6.72e-05	2.63e-04	2.03e-04	6.92e-08	0.717	-1.93e-06
svr ★	0.450	0.502	-0.052	0.075 [0.070, 0.079]	yes	2.24e-04	1.79e-04	5.02e-08	0.747	5.04e-05	2.64e-04	2.02e-04	6.96e-08	1.001	-1.54e-05
kernelridge	0.451	0.482	-0.031	0.054 [0.047, 0.061]	yes	2.29e-04	1.83e-04	5.22e-08	0.694	6.50e-05	2.64e-04	2.03e-04	6.95e-08	0.942	-1.23e-06
gbm	0.359	0.364	-0.005	-0.063 [-0.072, -0.053]	no	2.53e-04	2.03e-04	6.41e-08	0.408	9.38e-05	2.85e-04	2.20e-04	8.12e-08	0.535	1.31e-08
rf	0.463	0.480	-0.016	0.052 [0.046, 0.059]	yes	2.29e-04	1.84e-04	5.24e-08	0.473	6.82e-05	2.61e-04	2.01e-04	6.79e-08	0.657	-3.52e-08
extratrees	0.490	0.474	0.015	0.047 [0.040, 0.053]	yes	2.30e-04	1.85e-04	5.30e-08	0.459	7.29e-05	2.54e-04	1.96e-04	6.46e-08	0.596	-4.05e-21
mlp	0.463	0.481	-0.018	0.054 [0.047, 0.061]	yes	2.29e-04	1.84e-04	5.23e-08	0.504	6.89e-05	2.61e-04	2.01e-04	6.80e-08	0.700	-9.00e-07
gp	0.441	0.469	-0.027	0.042 [0.036, 0.047]	yes	2.31e-04	1.85e-04	5.35e-08	0.509	7.12e-05	2.66e-04	2.05e-04	7.07e-08	0.704	8.62e-08

best-by-test-R2 = svr (0.502) · linear 0.468 · persistence 0.427 · H=30s · 2728.3s. te.* = OOS test losses; tr.* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

horizon=30s Sigma=30s · horizon=30s (H=30s) — metrics: LE + AI

LE Sigma=30s | horizon=30s | metric=LE

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
linear	0.400	0.468	-0.068	0.046 [0.042, 0.050]	yes	2.26e-04	1.82e-04	5.09e-08	0.630	5.31e-05	2.67e-04	2.08e-04	7.15e-08	0.883	2.73e-16
ridge	0.400	0.468	-0.068	0.046 [0.042, 0.050]	yes	2.26e-04	1.82e-04	5.09e-08	0.630	5.31e-05	2.67e-04	2.08e-04	7.15e-08	0.883	-4.86e-17
lasso	0.397	0.456	-0.059	0.033 [0.030, 0.036]	yes	2.28e-04	1.84e-04	5.21e-08	0.570	5.72e-05	2.68e-04	2.09e-04	7.20e-08	0.799	-3.79e-20
elasticnet	0.385	0.437	-0.052	0.015 [0.013, 0.017]	yes	2.32e-04	1.87e-04	5.39e-08	0.537	6.12e-05	2.71e-04	2.10e-04	7.33e-08	0.748	-4.31e-20
huber	0.398	0.481	-0.083	0.058 [0.056, 0.061]	yes	2.23e-04	1.79e-04	4.97e-08	0.770	3.65e-05	2.68e-04	2.08e-04	7.18e-08	1.081	-1.60e-05
poly2	0.422	0.485	-0.063	0.062 [0.058, 0.067]	yes	2.22e-04	1.80e-04	4.93e-08	0.557	5.81e-05	2.62e-04	2.05e-04	6.89e-08	0.759	7.81e-15
poly3	0.421	0.476	-0.055	0.054 [0.050, 0.058]	yes	2.24e-04	1.81e-04	5.01e-08	0.521	6.15e-05	2.63e-04	2.05e-04	6.90e-08	0.825	1.43e-13
knn	0.452	0.460	-0.008	0.037 [0.031, 0.043]	yes	2.27e-04	1.84e-04	5.17e-08	0.505	6.66e-05	2.56e-04	1.99e-04	6.54e-08	0.704	-1.53e-06
svr ★	0.451	0.505	-0.055	0.083 [0.078, 0.087]	yes	2.18e-04	1.75e-04	4.74e-08	0.968	4.53e-05	2.56e-04	1.98e-04	6.55e-08	0.962	-1.48e-05
kernelridge	0.451	0.486	-0.034	0.063 [0.057, 0.069]	yes	2.22e-04	1.79e-04	4.93e-08	0.688	6.34e-05	2.56e-04	1.99e-04	6.54e-08	0.718	-1.18e-06
gbm	0.357	0.362	-0.005	-0.061 [-0.069, -0.051]	no	2.47e-04	2.00e-04	6.11e-08	0.395	9.35e-05	2.77e-04	2.16e-04	7.67e-08	0.523	8.70e-08
rf	0.464	0.480	-0.017	0.058 [0.050, 0.065]	yes	2.23e-04	1.81e-04	4.98e-08	0.478	6.65e-05	2.53e-04	1.97e-04	6.40e-08	0.648	-8.85e-10
extratrees	0.490	0.474	0.016	0.051 [0.044, 0.058]	yes	2.24e-04	1.82e-04	5.04e-08	0.461	7.09e-05	2.47e-04	1.92e-04	6.08e-08	0.588	1.86e-19
mlp	0.463	0.483	-0.019	0.060 [0.052, 0.067]	yes	2.23e-04	1.80e-04	4.95e-08	0.509	6.67e-05	2.53e-04	1.97e-04	6.40e-08	0.683	-1.75e-06
gp	0.440	0.489	-0.048	0.066 [0.062, 0.070]	yes	2.21e-04	1.79e-04	4.90e-08	0.546	6.26e-05	2.58e-04	2.01e-04	6.68e-08	0.759	-4.34e-06

best-by-test-R2 = svr (0.505) · linear 0.468 · persistence 0.423 · H=30s · 2676.5s. te.* = OOS test losses; tr.* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

AI Sigma=30s | horizon=30s | metric=AI

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
linear	0.398	0.465	-0.067	0.042 [0.038, 0.046]	yes	2.26e-04	1.82e-04	5.12e-08	0.639	5.28e-05	2.68e-04	2.09e-04	7.18e-08	0.886	-3.08e-16
ridge	0.398	0.465	-0.067	0.042 [0.038, 0.046]	yes	2.26e-04	1.82e-04	5.12e-08	0.639	5.28e-05	2.68e-04	2.09e-04	7.18e-08	0.886	-5.03e-17
lasso	0.394	0.453	-0.059	0.031 [0.027, 0.034]	yes	2.29e-04	1.84e-04	5.24e-08	0.573	5.76e-05	2.69e-04	2.09e-04	7.22e-08	0.801	-2.98e-20
elasticnet	0.384	0.436	-0.052	0.014 [0.011, 0.016]	yes	2.32e-04	1.87e-04	5.40e-08	0.535	6.23e-05	2.71e-04	2.11e-04	7.34e-08	0.747	-3.02e-20
huber	0.395	0.477	-0.082	0.054 [0.052, 0.057]	yes	2.24e-04	1.80e-04	5.01e-08	0.779	3.64e-05	2.69e-04	2.08e-04	7.21e-08	1.088	-1.59e-05
poly2	0.419	0.477	-0.058	0.055 [0.050, 0.060]	yes	2.24e-04	1.81e-04	5.00e-08	0.548	5.98e-05	2.63e-04	2.05e-04	6.93e-08	0.751	3.97e-15
poly3	0.419	0.467	-0.048	0.045 [0.040, 0.050]	yes	2.26e-04	1.83e-04	5.10e-08	0.513	6.49e-05	2.63e-04	2.05e-04	6.93e-08	0.798	4.99e-14

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
knn	0.450	0.462	-0.013	0.040 [0.035, 0.045]	yes	2.27e-04	1.83e-04	5.15e-08	0.500	6.61e-05	2.56e-04	1.99e-04	6.56e-08	0.708	-2.26e-06
svr ★	0.448	0.504	-0.056	0.081 [0.077, 0.086]	yes	2.18e-04	1.76e-04	4.75e-08	0.921	4.49e-05	2.56e-04	1.99e-04	6.58e-08	0.936	-1.49e-05
kernelridge	0.449	0.482	-0.033	0.060 [0.054, 0.066]	yes	2.23e-04	1.80e-04	4.96e-08	0.572	6.35e-05	2.56e-04	1.99e-04	6.57e-08	0.707	-1.16e-06
gbm	0.355	0.359	-0.004	-0.063 [-0.072, -0.053]	no	2.48e-04	2.00e-04	6.13e-08	0.396	9.33e-05	2.77e-04	2.16e-04	7.69e-08	0.525	8.92e-08
rf	0.461	0.478	-0.017	0.055 [0.048, 0.062]	yes	2.24e-04	1.81e-04	5.00e-08	0.466	6.73e-05	2.54e-04	1.98e-04	6.43e-08	0.648	-4.55e-09
extratrees	0.488	0.471	0.017	0.048 [0.040, 0.056]	yes	2.25e-04	1.83e-04	5.07e-08	0.451	7.24e-05	2.47e-04	1.92e-04	6.11e-08	0.588	1.83e-19
mlp	0.460	0.477	-0.017	0.055 [0.046, 0.063]	yes	2.24e-04	1.81e-04	5.01e-08	0.478	7.01e-05	2.54e-04	1.97e-04	6.44e-08	0.666	2.74e-07
gp	0.439	0.485	-0.046	0.063 [0.058, 0.068]	yes	2.22e-04	1.79e-04	4.93e-08	0.542	6.20e-05	2.59e-04	2.01e-04	6.68e-08	0.758	-4.47e-06

best-by-test-R2 = **svr** (0.504) · linear 0.465 · persistence 0.423 · H=30s · 2661.7s. te.* = OOS test losses; tr.* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

horizon=45sSigma=30s · horizon=45s (H=45s) — metrics: LE + AI

LESigma=30s | horizon=45s | metric=LE

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
linear	0.400	0.468	-0.068	0.046 [0.042, 0.050]	yes	2.26e-04	1.82e-04	5.09e-08	0.630	5.30e-05	2.67e-04	2.08e-04	7.15e-08	0.883	6.84e-16
ridge	0.400	0.468	-0.068	0.046 [0.042, 0.050]	yes	2.26e-04	1.82e-04	5.09e-08	0.630	5.30e-05	2.67e-04	2.08e-04	7.15e-08	0.883	-1.30e-17
lasso	0.397	0.455	-0.059	0.033 [0.030, 0.036]	yes	2.28e-04	1.84e-04	5.21e-08	0.570	5.72e-05	2.68e-04	2.09e-04	7.19e-08	0.799	1.47e-21
elasticnet	0.385	0.437	-0.052	0.015 [0.013, 0.017]	yes	2.32e-04	1.87e-04	5.38e-08	0.537	6.12e-05	2.71e-04	2.10e-04	7.33e-08	0.748	-7.37e-22
huber	0.398	0.481	-0.083	0.058 [0.056, 0.061]	yes	2.23e-04	1.79e-04	4.97e-08	0.770	3.65e-05	2.68e-04	2.08e-04	7.17e-08	1.081	-1.60e-05
poly2	0.422	0.485	-0.063	0.062 [0.058, 0.067]	yes	2.22e-04	1.80e-04	4.93e-08	0.557	5.81e-05	2.62e-04	2.05e-04	6.89e-08	0.759	6.41e-15
poly3	0.421	0.476	-0.055	0.054 [0.050, 0.058]	yes	2.24e-04	1.81e-04	5.01e-08	0.521	6.15e-05	2.63e-04	2.05e-04	6.90e-08	0.826	4.11e-13
knn	0.452	0.461	-0.009	0.039 [0.032, 0.045]	yes	2.27e-04	1.83e-04	5.16e-08	0.506	6.59e-05	2.56e-04	1.99e-04	6.53e-08	0.710	-1.95e-06
svr ★	0.450	0.495	-0.045	0.073 [0.066, 0.079]	yes	2.20e-04	1.77e-04	4.83e-08	0.739	5.01e-05	2.56e-04	1.99e-04	6.55e-08	0.939	-1.38e-05
kernelridge	0.450	0.476	-0.026	0.054 [0.044, 0.062]	yes	2.24e-04	1.80e-04	5.01e-08	1.080	6.31e-05	2.56e-04	1.99e-04	6.55e-08	0.925	1.32e-06
gbm	0.357	0.363	-0.005	-0.060 [-0.068, -0.050]	no	2.47e-04	1.99e-04	6.10e-08	0.394	9.33e-05	2.77e-04	2.16e-04	7.66e-08	0.523	-5.71e-08
rf	0.464	0.480	-0.017	0.058 [0.050, 0.065]	yes	2.23e-04	1.81e-04	4.97e-08	0.478	6.64e-05	2.53e-04	1.97e-04	6.39e-08	0.648	-7.57e-09
extratrees	0.491	0.474	0.016	0.052 [0.044, 0.058]	yes	2.24e-04	1.82e-04	5.03e-08	0.462	7.06e-05	2.46e-04	1.92e-04	6.07e-08	0.588	1.85e-19
mlp	0.463	0.488	-0.025	0.065 [0.058, 0.072]	yes	2.21e-04	1.79e-04	4.90e-08	0.506	6.31e-05	2.53e-04	1.97e-04	6.40e-08	0.700	-3.78e-06
gp	0.441	0.472	-0.032	0.050 [0.041, 0.057]	yes	2.25e-04	1.81e-04	5.05e-08	0.490	6.61e-05	2.58e-04	2.01e-04	6.67e-08	0.685	-9.91e-07

best-by-test-R2 = **svr** (0.495) · linear 0.468 · persistence 0.422 · H=45s · 2681.8s. te.* = OOS test losses; tr.* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

AISigma=30s | horizon=45s | metric=AI

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
linear	0.398	0.465	-0.067	0.042 [0.038, 0.047]	yes	2.26e-04	1.82e-04	5.12e-08	0.639	5.28e-05	2.68e-04	2.09e-04	7.18e-08	0.887	-7.61e-16
ridge	0.398	0.465	-0.067	0.042 [0.038, 0.047]	yes	2.26e-04	1.82e-04	5.12e-08	0.639	5.28e-05	2.68e-04	2.09e-04	7.18e-08	0.886	-1.36e-17
lasso	0.394	0.453	-0.059	0.031 [0.027, 0.034]	yes	2.29e-04	1.84e-04	5.24e-08	0.574	5.76e-05	2.69e-04	2.09e-04	7.22e-08	0.801	7.00e-21
elasticnet	0.384	0.436	-0.052	0.014 [0.011, 0.016]	yes	2.32e-04	1.87e-04	5.40e-08	0.535	6.23e-05	2.71e-04	2.11e-04	7.34e-08	0.747	-1.03e-20
huber	0.395	0.477	-0.082	0.054 [0.052, 0.057]	yes	2.24e-04	1.80e-04	5.01e-08	0.780	3.64e-05	2.69e-04	2.08e-04	7.21e-08	1.088	-1.59e-05
poly2	0.419	0.477	-0.058	0.055 [0.050, 0.060]	yes	2.24e-04	1.81e-04	5.00e-08	0.548	5.98e-05	2.63e-04	2.05e-04	6.92e-08	0.751	2.56e-15
poly3	0.419	0.467	-0.048	0.045 [0.040, 0.050]	yes	2.26e-04	1.83e-04	5.10e-08	0.514	6.48e-05	2.63e-04	2.05e-04	6.93e-08	0.798	1.11e-13
knn	0.450	0.463	-0.013	0.041 [0.035, 0.046]	yes	2.27e-04	1.83e-04	5.14e-08	0.501	6.55e-05	2.56e-04	1.99e-04	6.56e-08	0.713	-2.64e-06
svr ★	0.448	0.497	-0.049	0.075 [0.070, 0.079]	yes	2.19e-04	1.77e-04	4.81e-08	0.706	4.93e-05	2.56e-04	1.99e-04	6.57e-08	0.920	-1.35e-05
kernelridge	0.449	0.478	-0.029	0.056 [0.049, 0.062]	yes	2.23e-04	1.80e-04	4.99e-08	0.740	6.30e-05	2.56e-04	1.99e-04	6.56e-08	0.746	1.54e-06
gbm	0.355	0.360	-0.004	-0.063 [-0.072, -0.053]	no	2.48e-04	2.00e-04	6.13e-08	0.397	9.30e-05	2.77e-04	2.16e-04	7.69e-08	0.526	-5.19e-08
rf	0.461	0.477	-0.016	0.055 [0.048, 0.062]	yes	2.24e-04	1.81e-04	5.01e-08	0.466	6.75e-05	2.54e-04	1.98e-04	6.43e-08	0.648	-6.11e-09
extratrees	0.487	0.470	0.017	0.048 [0.040, 0.056]	yes	2.25e-04	1.83e-04	5.07e-08	0.449	7.28e-05	2.47e-04	1.92e-04	6.11e-08	0.587	1.88e-19
mlp	0.460	0.482	-0.022	0.060 [0.052, 0.067]	yes	2.23e-04	1.80e-04	4.95e-08	0.476	6.80e-05	2.54e-04	1.98e-04	6.44e-08	0.656	1.33e-06
gp	0.439	0.470	-0.031	0.048 [0.039, 0.055]	yes	2.25e-04	1.81e-04	5.07e-08	0.502	6.49e-05	2.59e-04	2.01e-04	6.68e-08	0.690	-1.36e-06

best-by-test-R2 = **svr** (0.497) · linear 0.465 · persistence 0.422 · H=45s · 2656.5s. te.* = OOS test losses; tr.* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

horizon=60sSigma=30s · horizon=60s (H=60s) — metrics: LE + AI

LESigma=30s | horizon=60s | metric=LE

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
linear	0.454	0.533	-0.079	0.048 [0.044, 0.053]	yes	2.66e-04	2.10e-04	7.06e-08	0.410	8.38e-05	3.14e-04	2.42e-04	9.88e-08	0.524	2.09e-16
ridge	0.454	0.533	-0.079	0.048 [0.044, 0.053]	yes	2.66e-04	2.10e-04	7.06e-08	0.410	8.38e-05	3.14e-04	2.42e-04	9.88e-08	0.524	-1.60e-17
lasso	0.452	0.522	-0.070	0.037 [0.033, 0.041]	yes	2.69e-04	2.13e-04	7.23e-08	0.401	8.79e-05	3.15e-04	2.42e-04	9.93e-08	0.501	7.83e-21
elasticnet	0.444	0.507	-0.063	0.022 [0.019, 0.025]	yes	2.73e-04	2.16e-04	7.46e-08	0.397	9.21e-05	3.17e-04	2.43e-04	1.01e-07	0.487	-3.70e-20
huber	0.452	0.551	-0.099	0.066 [0.063, 0.068]	yes	2.61e-04	2.06e-04	6.79e-08	0.419	6.66e-05	3.15e-04	2.41e-04	9.93e-08	0.555	-1.97e-05
poly2	0.479	0.547	-0.068	0.062 [0.055, 0.069]	yes	2.62e-04	2.07e-04	6.85e-08	0.399	9.19e-05	3.07e-04	2.36e-04	9.44e-08	0.479	-3.36e-15
poly3	0.479	0.535	-0.057	0.050 [0.044, 0.057]	yes	2.65e-04	2.10e-04	7.02e-08	0.389	9.73e-05	3.07e-04	2.36e-04	9.44e-08	0.486	2.91e-13
knn	0.516	0.518	-0.002	0.033 [0.026, 0.041]	yes	2.70e-04	2.14e-04	7.28e-08	0.373	1.05e-04	2.96e-04	2.26e-04	8.76e-08	0.421	-2.40e-06
svr ★	0.514	0.558	-0.044	0.073 [0.067, 0.080]	yes	2.58e-04	2.04e-04	6.68e-08	0.365	8.90e-05	2.97e-04	2.26e-04	8.80e-08	0.428	-1.33e-05
kernelridge	0.514	0.534	-0.020	0.049 [0.039, 0.058]	yes	2.65e-04	2.10e-04	7.04e-08	0.535	1.02e-04	2.97e-04	2.26e-04	8.80e-08	0.522	3.17e-07
gbm	0.406	0.400	0.006	-0.085 [-0.093, -0.074]	no	3.01e-04	2.39e-04	9.06e-08	0.399	1.38e-04	3.28e-04	2.50e-04	1.08e-07	0.402	2.04e-07
rf	0.535	0.537	-0.002	0.052 [0.042, 0.062]	yes	2.65e-04	2.11e-04	7.00e-08	0.371	1.03e-04	2.90e-04	2.22e-04	8.43e-08	0.400	2.29e-08
extratrees	0.553	0.531	0.022	0.046 [0.036, 0.055]	yes	2.66e-04	2.12e-04	7.09e-08	0.368	1.08e-04	2.84e-04	2.17e-04	8.09e-08	0.372	5.98e-20
mlp	0.527	0.542	-0.015	0.058 [0.047, 0.067]	yes	2.63e-04	2.10e-04	6.92e-08	0.372	1.03e-04	2.93e-04	2.23e-04	8.56e-08	0.417	-3.04e-06
gp	0.502	0.525	-0.023	0.040 [0.031, 0.049]	yes	2.68e-04	2.11e-04	7.18e-08	0.370	1.07e-04	3.00e-04	2.29e-04	9.01e-08	0.418	-4.53e-07

best-by-test-R2 = **svr** (0.558) · linear 0.533 · persistence 0.485 · H=60s · 2688.0s. te.* = OOS test losses; tr.* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

AISigma=30s | horizon=60s | metric=AI

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
linear	0.451	0.529	-0.078	0.044 [0.039, 0.049]	yes	2.67e-04	2.11e-04	7.12e-08	0.415	8.38e-05	3.15e-04	2.42e-04	9.94e-08	0.527	-4.34e-16

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
ridge	0.451	0.529	-0.078	0.044 [0.039, 0.049]	yes	2.67e-04	2.11e-04	7.12e-08	0.415	8.38e-05	3.15e-04	2.42e-04	9.94e-08	0.527	-1.64e-17
lasso	0.449	0.518	-0.070	0.034 [0.029, 0.038]	yes	2.70e-04	2.14e-04	7.28e-08	0.404	8.86e-05	3.16e-04	2.42e-04	9.98e-08	0.503	3.13e-20
elasticnet	0.442	0.504	-0.062	0.019 [0.015, 0.023]	yes	2.74e-04	2.17e-04	7.50e-08	0.399	9.35e-05	3.18e-04	2.44e-04	1.01e-07	0.487	-3.56e-20
huber	0.448	0.546	-0.097	0.061 [0.058, 0.063]	yes	2.62e-04	2.07e-04	6.87e-08	0.425	6.69e-05	3.16e-04	2.41e-04	9.99e-08	0.560	-1.96e-05
poly2	0.475	0.537	-0.062	0.052 [0.045, 0.060]	yes	2.64e-04	2.09e-04	7.00e-08	0.400	9.43e-05	3.08e-04	2.36e-04	9.50e-08	0.478	-1.21e-15
poly3	0.477	0.524	-0.047	0.039 [0.031, 0.047]	yes	2.68e-04	2.13e-04	7.19e-08	0.392	1.02e-04	3.08e-04	2.36e-04	9.48e-08	0.477	-1.14e-13
knn	0.515	0.521	-0.006	0.036 [0.029, 0.043]	yes	2.69e-04	2.14e-04	7.24e-08	0.372	1.04e-04	2.96e-04	2.26e-04	8.79e-08	0.424	-3.49e-06
svr ★	0.512	0.560	-0.048	0.075 [0.070, 0.081]	yes	2.58e-04	2.04e-04	6.65e-08	0.360	8.92e-05	2.97e-04	2.26e-04	8.84e-08	0.430	-1.32e-05
kernelridge	0.512	0.535	-0.023	0.050 [0.042, 0.057]	yes	2.65e-04	2.10e-04	7.03e-08	0.563	1.03e-04	2.97e-04	2.27e-04	8.84e-08	0.504	3.29e-07
gbm	0.404	0.401	0.003	-0.084 [-0.093, -0.074]	no	3.01e-04	2.39e-04	9.06e-08	0.399	1.37e-04	3.29e-04	2.51e-04	1.08e-07	0.404	2.03e-07
rf	0.523	0.533	-0.010	0.048 [0.038, 0.059]	yes	2.66e-04	2.12e-04	7.05e-08	0.366	1.06e-04	2.94e-04	2.25e-04	8.63e-08	0.406	2.73e-08
extratrees	0.551	0.526	0.025	0.041 [0.030, 0.052]	yes	2.68e-04	2.14e-04	7.17e-08	0.367	1.11e-04	2.85e-04	2.18e-04	8.14e-08	0.373	5.86e-20
mlp	0.525	0.532	-0.007	0.047 [0.035, 0.058]	yes	2.66e-04	2.12e-04	7.08e-08	0.369	1.11e-04	2.93e-04	2.24e-04	8.60e-08	0.413	-1.57e-06
gp	0.501	0.522	-0.021	0.037 [0.029, 0.045]	yes	2.69e-04	2.12e-04	7.22e-08	0.367	1.07e-04	3.01e-04	2.29e-04	9.03e-08	0.416	-2.46e-07

best-by-test-R2 = **svr** (0.560) · linear 0.529 · persistence 0.485 · H=60s · 2684.7s. te.* = OOS test losses; tr.* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

horizon=2mSigma=30s · horizon=2m (H=120s) — metrics: LE + AI

LESigma=30s | horizon=2m | metric=LE

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
linear	0.505	0.587	-0.082	0.051 [0.046, 0.057]	yes	3.25e-04	2.57e-04	1.06e-07	0.332	1.28e-04	3.83e-04	2.92e-04	1.47e-07	0.381	9.53e-16
ridge	0.505	0.587	-0.082	0.051 [0.046, 0.057]	yes	3.25e-04	2.58e-04	1.06e-07	0.332	1.28e-04	3.83e-04	2.92e-04	1.47e-07	0.381	1.78e-17
lasso	0.504	0.578	-0.074	0.042 [0.037, 0.047]	yes	3.29e-04	2.60e-04	1.08e-07	0.333	1.32e-04	3.84e-04	2.92e-04	1.47e-07	0.374	1.02e-19
elasticnet	0.499	0.566	-0.067	0.031 [0.027, 0.035]	yes	3.33e-04	2.64e-04	1.11e-07	0.335	1.36e-04	3.85e-04	2.93e-04	1.48e-07	0.369	7.93e-20
huber	0.501	0.610	-0.109	0.075 [0.072, 0.078]	yes	3.16e-04	2.48e-04	9.98e-08	0.325	1.04e-04	3.84e-04	2.90e-04	1.48e-07	0.393	-2.94e-05
poly2	0.534	0.600	-0.065	0.064 [0.055, 0.073]	yes	3.20e-04	2.54e-04	1.03e-07	0.334	1.40e-04	3.72e-04	2.82e-04	1.38e-07	0.357	9.22e-15
poly3	0.534	0.584	-0.050	0.049 [0.040, 0.057]	yes	3.26e-04	2.59e-04	1.06e-07	0.333	1.48e-04	3.72e-04	2.83e-04	1.38e-07	0.354	4.80e-13
knn	0.580	0.570	0.010	0.034 [0.026, 0.043]	yes	3.32e-04	2.63e-04	1.10e-07	0.321	1.57e-04	3.53e-04	2.65e-04	1.25e-07	0.301	-3.56e-06
svr ★	0.575	0.629	-0.055	0.094 [0.087, 0.101]	yes	3.08e-04	2.43e-04	9.49e-08	0.309	1.20e-04	3.55e-04	2.65e-04	1.26e-07	0.321	-2.75e-05
kernelridge	0.575	0.593	-0.019	0.058 [0.048, 0.067]	yes	3.23e-04	2.56e-04	1.04e-07	0.395	1.50e-04	3.55e-04	2.67e-04	1.26e-07	0.367	-1.87e-06
gbm	0.455	0.436	0.019	-0.100 [-0.108, -0.089]	no	3.80e-04	3.06e-04	1.44e-07	0.387	2.02e-04	4.02e-04	3.04e-04	1.62e-07	0.336	-1.22e-07
rf	0.619	0.569	0.050	0.033 [0.018, 0.047]	yes	3.32e-04	2.66e-04	1.10e-07	0.328	1.64e-04	3.36e-04	2.53e-04	1.13e-07	0.277	6.40e-08
extratrees	0.615	0.575	0.040	0.039 [0.027, 0.050]	yes	3.30e-04	2.65e-04	1.09e-07	0.325	1.62e-04	3.38e-04	2.54e-04	1.14e-07	0.273	3.01e-21
mlp	0.592	0.590	0.002	0.054 [0.039, 0.066]	yes	3.24e-04	2.59e-04	1.05e-07	0.316	1.56e-04	3.48e-04	2.62e-04	1.21e-07	0.293	1.10e-06
gp	0.560	0.584	-0.024	0.048 [0.035, 0.059]	yes	3.27e-04	2.57e-04	1.07e-07	0.320	1.49e-04	3.61e-04	2.71e-04	1.31e-07	0.316	-4.69e-06

best-by-test-R2 = **svr** (0.629) · linear 0.587 · persistence 0.536 · H=120s · 2735.9s. te.* = OOS test losses; tr.* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

AISigma=30s | horizon=2m | metric=AI

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
linear	0.502	0.583	-0.081	0.047 [0.041, 0.053]	yes	3.27e-04	2.59e-04	1.07e-07	0.336	1.28e-04	3.84e-04	2.93e-04	1.48e-07	0.384	2.66e-16
ridge	0.502	0.582	-0.081	0.047 [0.041, 0.053]	yes	3.27e-04	2.59e-04	1.07e-07	0.336	1.28e-04	3.84e-04	2.93e-04	1.48e-07	0.384	1.79e-17
lasso	0.500	0.574	-0.073	0.038 [0.032, 0.043]	yes	3.30e-04	2.62e-04	1.09e-07	0.336	1.33e-04	3.85e-04	2.93e-04	1.48e-07	0.376	8.99e-20
elasticnet	0.496	0.563	-0.066	0.027 [0.022, 0.032]	yes	3.35e-04	2.65e-04	1.12e-07	0.337	1.38e-04	3.86e-04	2.94e-04	1.49e-07	0.371	2.41e-20
huber	0.498	0.605	-0.107	0.069 [0.066, 0.072]	yes	3.18e-04	2.50e-04	1.01e-07	0.330	1.04e-04	3.86e-04	2.91e-04	1.49e-07	0.397	-2.94e-05
poly2	0.531	0.588	-0.057	0.052 [0.042, 0.062]	yes	3.25e-04	2.58e-04	1.06e-07	0.336	1.44e-04	3.73e-04	2.83e-04	1.39e-07	0.358	-4.27e-15
poly3	0.533	0.571	-0.039	0.036 [0.025, 0.046]	yes	3.31e-04	2.64e-04	1.10e-07	0.339	1.55e-04	3.72e-04	2.83e-04	1.39e-07	0.374	-4.99e-13
knn	0.578	0.573	0.005	0.037 [0.028, 0.045]	yes	3.31e-04	2.63e-04	1.09e-07	0.321	1.57e-04	3.54e-04	2.66e-04	1.25e-07	0.303	-5.13e-06
svr ★	0.571	0.624	-0.053	0.089 [0.082, 0.096]	yes	3.10e-04	2.44e-04	9.62e-08	0.302	1.24e-04	3.56e-04	2.66e-04	1.27e-07	0.322	-2.80e-05
kernelridge	0.572	0.587	-0.015	0.051 [0.041, 0.060]	yes	3.25e-04	2.57e-04	1.06e-07	0.398	1.53e-04	3.56e-04	2.67e-04	1.27e-07	0.386	-1.80e-06
gbm	0.451	0.438	0.013	-0.097 [-0.106, -0.088]	no	3.79e-04	3.05e-04	1.44e-07	0.387	2.00e-04	4.03e-04	3.05e-04	1.63e-07	0.339	-1.08e-07
rf	0.616	0.552	0.063	0.017 [0.000, 0.033]	yes	3.39e-04	2.72e-04	1.15e-07	0.334	1.74e-04	3.37e-04	2.54e-04	1.14e-07	0.279	7.08e-08
extratrees	0.613	0.568	0.045	0.032 [0.017, 0.046]	yes	3.33e-04	2.67e-04	1.11e-07	0.328	1.68e-04	3.39e-04	2.55e-04	1.15e-07	0.274	2.23e-20
mlp	0.588	0.575	0.013	0.039 [0.024, 0.055]	yes	3.30e-04	2.65e-04	1.09e-07	0.322	1.70e-04	3.49e-04	2.63e-04	1.22e-07	0.293	2.73e-06
gp	0.559	0.573	-0.014	0.038 [0.026, 0.049]	yes	3.31e-04	2.59e-04	1.09e-07	0.326	1.51e-04	3.61e-04	2.71e-04	1.31e-07	0.317	-4.47e-06

best-by-test-R2 = **svr** (0.624) · linear 0.583 · persistence 0.536 · H=120s · 2732.6s. te.* = OOS test losses; tr.* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

horizon=5mSigma=30s · horizon=5m (H=300s) — metrics: LE + AI

LESigma=30s | horizon=5m | metric=LE

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
linear	0.545	0.627	-0.082	0.055 [0.049, 0.061]	yes	4.61e-04	3.68e-04	2.13e-07	0.256	2.16e-04	5.38e-04	4.08e-04	2.90e-07	0.285	4.82e-16
ridge	0.545	0.627	-0.082	0.055 [0.049, 0.061]	yes	4.61e-04	3.68e-04	2.13e-07	0.256	2.16e-04	5.38e-04	4.08e-04	2.90e-07	0.285	7.10e-17
lasso	0.544	0.620	-0.076	0.049 [0.043, 0.054]	yes	4.65e-04	3.71e-04	2.17e-07	0.258	2.20e-04	5.39e-04	4.09e-04	2.90e-07	0.283	-2.99e-20
elasticnet	0.542	0.613	-0.071	0.041 [0.035, 0.046]	yes	4.70e-04	3.75e-04	2.21e-07	0.260	2.24e-04	5.40e-04	4.09e-04	2.91e-07	0.281	-4.49e-20
huber	0.539	0.657	-0.118	0.086 [0.082, 0.089]	yes	4.42e-04	3.49e-04	1.95e-07	0.245	1.75e-04	5.41e-04	4.04e-04	2.93e-07	0.294	-5.14e-05
poly2	0.578	0.638	-0.059	0.066 [0.056, 0.076]	yes	4.55e-04	3.65e-04	2.07e-07	0.261	2.35e-04	5.18e-04	3.90e-04	2.68e-07	0.268	-3.64e-15
poly3	0.578	0.619	-0.041	0.047 [0.037, 0.056]	yes	4.66e-04	3.74e-04	2.17e-07	0.267	2.49e-04	5.18e-04	3.91e-04	2.69e-07	0.278	8.78e-13
knn	0.633	0.604	0.029	0.032 [0.022, 0.042]	yes	4.75e-04	3.83e-04	2.26e-07	0.256	2.63e-04	4.83e-04	3.59e-04	2.33e-07	0.216	-3.87e-06
svr ★	0.626	0.682	-0.056	0.110 [0.102, 0.118]	yes	4.26e-04	3.38e-04	1.81e-07	0.226	1.98e-04	4.87e-04	3.59e-04	2.38e-07	0.229	-4.61e-05
kernelridge	0.626	0.622	0.005	0.050 [0.039, 0.061]	yes	4.64e-04	3.73e-04	2.16e-07	0.405	2.59e-04	4.88e-04	3.63e-04	2.38e-07	0.308	1.81e-06
gbm	0.532	0.507	0.025	-0.064 [-0.072, -0.055]	no	5.30e-04	4.34e-04	2.81e-07	0.313	3.12e-04	5.45e-04	4.10e-04	2.97e-07	0.252	1.95e-07
rf	0.671	0.590	0.081	0.018 [-0.000, 0.034]	no	4.83e-04	3.95e-04	2.34e-07	0.268	2.82e-04	4.57e-04	3.40e-04	2.09e-07	0.197	7.33e-08
extratrees	0.667	0.603	0.064	0.032 [0.017, 0.044]	yes	4.75e-04	3.88e-04	2.26e-07	0.263	2.73e-04	4.60e-04	3.41e-04	2.11e-07	0.194	3.33e-20
mlp	0.644	0.624	0.020	0.053 [0.035, 0.067]	yes	4.63e-04	3.75e-04	2.14e-07	0.249	2.59e-04	4.76e-04	3.53e-04	2.26e-07	0.212	-2.50e-06
gp	0.615	0.629	-0.014	0.057 [0.045, 0.067]	yes	4.60e-04	3.70e-04	2.12e-07	0.249	2.54e-04	4.95e-04	3.67e-04	2.45e-07	0.228	-7.84e-06

best-by-test-R2 = **svr** (0.682) · linear 0.627 · persistence 0.572 · H=300s · 2655.5s. te.* = OOS test losses; tr.* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

AI Sigma=30s | horizon=5m | metric=AI

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
linear	0.541	0.622	-0.080	0.050 [0.043, 0.057]	yes	4.64e-04	3.71e-04	2.16e-07	0.260	2.17e-04	5.40e-04	4.10e-04	2.92e-07	0.288	-2.76e-16
ridge	0.541	0.622	-0.080	0.050 [0.043, 0.057]	yes	4.64e-04	3.71e-04	2.16e-07	0.260	2.17e-04	5.40e-04	4.10e-04	2.92e-07	0.288	7.31e-17
lasso	0.541	0.615	-0.075	0.044 [0.037, 0.050]	yes	4.68e-04	3.74e-04	2.19e-07	0.261	2.22e-04	5.41e-04	4.10e-04	2.92e-07	0.285	-2.42e-20
elasticnet	0.539	0.608	-0.069	0.037 [0.030, 0.043]	yes	4.73e-04	3.77e-04	2.23e-07	0.263	2.27e-04	5.42e-04	4.11e-04	2.93e-07	0.283	2.71e-20
huber	0.536	0.651	-0.115	0.079 [0.075, 0.083]	yes	4.46e-04	3.52e-04	1.99e-07	0.249	1.75e-04	5.43e-04	4.06e-04	2.95e-07	0.298	-5.16e-05
poly2	0.574	0.624	-0.049	0.052 [0.041, 0.063]	yes	4.63e-04	3.71e-04	2.14e-07	0.266	2.41e-04	5.20e-04	3.92e-04	2.71e-07	0.269	-1.46e-14
poly3	0.577	0.604	-0.027	0.032 [0.019, 0.044]	yes	4.75e-04	3.83e-04	2.26e-07	0.274	2.60e-04	5.19e-04	3.91e-04	2.69e-07	0.308	7.34e-13
knn	0.631	0.606	0.025	0.034 [0.023, 0.044]	yes	4.74e-04	3.83e-04	2.25e-07	0.257	2.64e-04	4.85e-04	3.60e-04	2.35e-07	0.218	-6.20e-06
svr ★	0.624	0.671	-0.048	0.100 [0.091, 0.108]	yes	4.33e-04	3.44e-04	1.87e-07	0.230	2.06e-04	4.89e-04	3.60e-04	2.39e-07	0.231	-4.67e-05
kernelridge	0.624	0.613	0.011	0.041 [0.030, 0.052]	yes	4.70e-04	3.78e-04	2.21e-07	0.314	2.65e-04	4.89e-04	3.64e-04	2.39e-07	0.267	2.02e-06
gbm	0.527	0.502	0.025	-0.069 [-0.077, -0.061]	no	5.33e-04	4.36e-04	2.84e-07	0.315	3.12e-04	5.48e-04	4.13e-04	3.01e-07	0.257	1.96e-07
rf	0.668	0.572	0.096	0.001 [-0.021, 0.020]	no	4.94e-04	4.04e-04	2.44e-07	0.275	2.94e-04	4.59e-04	3.41e-04	2.11e-07	0.199	9.57e-08
extratrees	0.664	0.591	0.073	0.020 [0.002, 0.036]	yes	4.83e-04	3.95e-04	2.33e-07	0.269	2.85e-04	4.62e-04	3.43e-04	2.13e-07	0.196	3.74e-20
mlp	0.641	0.591	0.049	0.020 [-0.001, 0.039]	no	4.83e-04	3.94e-04	2.33e-07	0.266	2.89e-04	4.78e-04	3.55e-04	2.28e-07	0.214	1.10e-06
gp	0.616	0.619	-0.003	0.048 [0.035, 0.058]	yes	4.66e-04	3.74e-04	2.17e-07	0.252	2.58e-04	4.94e-04	3.67e-04	2.44e-07	0.229	-6.72e-06

best-by-test-R2 = **svr** (0.671) · linear 0.622 · persistence 0.572 · H=300s · 2636.2s. te.* = OOS test losses; tr.* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

horizon=10m Sigma=30s · horizon=10m (H=600s) — metrics: LE + AI

LE Sigma=30s | horizon=10m | metric=LE

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
linear	0.544	0.630	-0.087	0.058 [0.051, 0.064]	yes	6.34e-04	5.09e-04	4.02e-07	0.227	3.20e-04	7.36e-04	5.56e-04	5.42e-07	0.252	1.31e-16
ridge	0.544	0.630	-0.087	0.058 [0.051, 0.064]	yes	6.34e-04	5.09e-04	4.02e-07	0.227	3.20e-04	7.36e-04	5.56e-04	5.42e-07	0.252	1.19e-16
lasso	0.543	0.625	-0.082	0.053 [0.046, 0.059]	yes	6.39e-04	5.13e-04	4.08e-07	0.228	3.25e-04	7.37e-04	5.56e-04	5.43e-07	0.250	4.29e-19
elasticnet	0.542	0.620	-0.078	0.047 [0.041, 0.053]	yes	6.43e-04	5.17e-04	4.14e-07	0.230	3.29e-04	7.38e-04	5.57e-04	5.44e-07	0.249	2.33e-19
huber	0.537	0.666	-0.130	0.094 [0.090, 0.098]	yes	6.03e-04	4.77e-04	3.63e-07	0.213	2.58e-04	7.42e-04	5.49e-04	5.51e-07	0.260	-7.78e-05
poly2	0.579	0.638	-0.060	0.066 [0.055, 0.077]	yes	6.27e-04	5.07e-04	3.94e-07	0.232	3.48e-04	7.07e-04	5.30e-04	5.00e-07	0.237	1.77e-14
poly3	0.578	0.617	-0.039	0.044 [0.033, 0.054]	yes	6.46e-04	5.22e-04	4.17e-07	0.240	3.68e-04	7.08e-04	5.31e-04	5.01e-07	0.257	3.13e-13
knn	0.638	0.600	0.038	0.027 [0.016, 0.037]	yes	6.60e-04	5.37e-04	4.36e-07	0.232	3.89e-04	6.56e-04	4.82e-04	4.30e-07	0.188	-4.00e-06
svr ★	0.629	0.682	-0.053	0.110 [0.100, 0.119]	yes	5.88e-04	4.68e-04	3.46e-07	0.200	2.88e-04	6.64e-04	4.81e-04	4.40e-07	0.201	-7.27e-05
kernelridge	0.629	0.623	0.006	0.050 [0.037, 0.062]	yes	6.41e-04	5.20e-04	4.11e-07	0.400	3.74e-04	6.64e-04	4.88e-04	4.41e-07	0.263	7.79e-07
gbm	0.568	0.543	0.025	-0.030 [-0.037, -0.020]	no	7.05e-04	5.84e-04	4.97e-07	0.270	4.34e-04	7.17e-04	5.35e-04	5.14e-07	0.213	-1.40e-07
rf	0.690	0.564	0.126	-0.008 [-0.032, 0.012]	no	6.89e-04	5.67e-04	4.74e-07	0.251	4.27e-04	6.06e-04	4.44e-04	3.68e-07	0.161	3.28e-07
extratrees	0.673	0.594	0.079	0.022 [0.004, 0.035]	yes	6.65e-04	5.48e-04	4.42e-07	0.241	4.06e-04	6.23e-04	4.57e-04	3.89e-07	0.168	1.52e-19
mlp	0.647	0.616	0.031	0.044 [0.024, 0.059]	yes	6.46e-04	5.31e-04	4.18e-07	0.227	3.94e-04	6.47e-04	4.75e-04	4.19e-07	0.185	-2.79e-06
gp	0.614	0.582	0.032	0.009 [-0.008, 0.025]	no	6.75e-04	5.45e-04	4.56e-07	0.239	4.00e-04	6.77e-04	4.96e-04	4.59e-07	0.197	1.30e-06

best-by-test-R2 = **svr** (0.682) · linear 0.630 · persistence 0.573 · H=600s · 2654.0s. te.* = OOS test losses; tr.* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

AI Sigma=30s | horizon=10m | metric=AI

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
linear	0.540	0.625	-0.086	0.053 [0.044, 0.060]	yes	6.39e-04	5.13e-04	4.08e-07	0.230	3.23e-04	7.39e-04	5.59e-04	5.47e-07	0.255	5.78e-16
ridge	0.540	0.625	-0.086	0.053 [0.044, 0.060]	yes	6.39e-04	5.13e-04	4.08e-07	0.230	3.23e-04	7.39e-04	5.59e-04	5.47e-07	0.255	1.23e-16
lasso	0.539	0.621	-0.081	0.048 [0.040, 0.055]	yes	6.43e-04	5.16e-04	4.13e-07	0.231	3.28e-04	7.40e-04	5.59e-04	5.47e-07	0.253	1.57e-19
elasticnet	0.538	0.615	-0.077	0.043 [0.035, 0.050]	yes	6.47e-04	5.20e-04	4.19e-07	0.233	3.33e-04	7.41e-04	5.60e-04	5.48e-07	0.252	3.80e-19
huber	0.533	0.660	-0.127	0.087 [0.083, 0.091]	yes	6.08e-04	4.81e-04	3.70e-07	0.217	2.60e-04	7.45e-04	5.52e-04	5.55e-07	0.265	-7.83e-05
poly2	0.574	0.624	-0.049	0.051 [-0.039, 0.063]	yes	6.40e-04	5.17e-04	4.10e-07	0.238	3.56e-04	7.11e-04	5.33e-04	5.06e-07	0.239	3.18e-14
poly3	0.577	0.601	-0.024	0.028 [0.014, 0.041]	yes	6.59e-04	5.35e-04	4.34e-07	0.259	3.84e-04	7.09e-04	5.30e-04	5.02e-07	0.275	1.06e-12
knn	0.635	0.600	0.035	0.027 [0.013, 0.039]	yes	6.60e-04	5.39e-04	4.36e-07	0.233	3.93e-04	6.58e-04	4.84e-04	4.33e-07	0.190	-7.49e-06
svr ★	0.627	0.674	-0.046	0.101 [0.090, 0.111]	yes	5.96e-04	4.76e-04	3.55e-07	0.204	3.02e-04	6.65e-04	4.82e-04	4.43e-07	0.203	-7.19e-05
kernelridge	0.627	0.616	0.012	0.043 [0.031, 0.054]	yes	6.47e-04	5.25e-04	4.18e-07	0.427	3.84e-04	6.65e-04	4.90e-04	4.43e-07	0.246	1.32e-06
gbm	0.563	0.537	0.026	-0.035 [-0.044, -0.025]	no	7.10e-04	5.88e-04	5.04e-07	0.273	4.37e-04	7.21e-04	5.39e-04	5.19e-07	0.216	-1.54e-07
rf	0.674	0.556	0.118	-0.017 [-0.041, 0.006]	no	6.95e-04	5.73e-04	4.83e-07	0.254	4.40e-04	6.23e-04	4.57e-04	3.88e-07	0.171	1.75e-07
extratrees	0.670	0.579	0.091	0.007 [-0.014, 0.026]	no	6.77e-04	5.60e-04	4.58e-07	0.248	4.25e-04	6.26e-04	4.59e-04	3.92e-07	0.169	5.76e-21
mlp	0.645	0.586	0.059	0.013 [-0.010, 0.035]	no	6.72e-04	5.52e-04	4.51e-07	0.240	4.23e-04	6.50e-04	4.77e-04	4.22e-07	0.187	-4.06e-06
gp	0.613	0.582	0.032	0.009 [-0.004, 0.021]	no	6.75e-04	5.49e-04	4.55e-07	0.241	4.11e-04	6.78e-04	4.97e-04	4.59e-07	0.198	2.59e-06

best-by-test-R2 = **svr** (0.674) · linear 0.625 · persistence 0.573 · H=600s · 2588.6s. te.* = OOS test losses; tr.* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

horizon=15m Sigma=30s · horizon=15m (H=900s) — metrics: LE + AI

LE Sigma=30s | horizon=15m | metric=LE

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
linear	0.533	0.625	-0.092	0.060 [0.053, 0.067]	yes	7.75e-04	6.25e-04	6.01e-07	0.219	4.06e-04	8.98e-04	6.75e-04	8.06e-07	0.242	9.28e-16
ridge	0.533	0.625	-0.092	0.060 [0.053, 0.067]	yes	7.75e-04	6.25e-04	6.01e-07	0.219	4.06e-04	8.98e-04	6.75e-04	8.06e-07	0.242	-1.70e-16
lasso	0.533	0.621	-0.088	0.056 [0.049, 0.063]	yes	7.80e-04	6.29e-04	6.08e-07	0.220	4.10e-04	8.98e-04	6.76e-04	8.06e-07	0.241	-3.79e-20
elasticnet	0.532	0.616	-0.084	0.051 [0.044, 0.058]	yes	7.84e-04	6.32e-04	6.15e-07	0.222	4.14e-04	8.99e-04	6.76e-04	8.08e-07	0.240	1.40e-19
huber	0.525	0.665	-0.140	0.100 [0.096, 0.104]	yes	7.32e-04	5.81e-04	5.36e-07	0.203	3.25e-04	9.05e-04	6.66e-04	8.19e-07	0.252	-1.00e-04
poly2	0.568	0.630	-0.062	0.065 [0.053, 0.077]	yes	7.70e-04	6.24e-04	5.93e-07	0.225	4.38e-04	8.63e-04	6.44e-04	7.45e-07	0.228	6.62e-15
poly3	0.568	0.607	-0.039	0.042 [0.031, 0.052]	yes	7.94e-04	6.44e-04	6.30e-07	0.233	4.64e-04	8.63e-04	6.44e-04	7.45e-07	0.248	-7.27e-13
knn	0.628	0.591	0.037	0.026 [0.015, 0.037]	yes	8.09e-04	6.62e-04	6.55e-07	0.226	4.89e-04	8.01e-04	5.85e-04	6.41e-07	0.182	-4.07e-06
svr ★	0.619	0.678	-0.059	0.113 [0.103, 0.123]	yes	7.18e-04	5.76e-04	5.16e-07	0.194	3.67e-04	8.11e-04	5.82e-04	6.57e-07	0.195	-9.32e-05
kernelridge	0.619	0.610	0.010	0.045 [0.031, 0.056]	yes	7.91e-04	6.47e-04	6.26e-07	0.272	4.78e-04	8.10e-04	5.92e-04	6.57e-07	0.222	3.78e-06

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
gbm	0.571	0.546	0.026	-0.019 [-0.027, -0.009]	no	8.53e-04	7.09e-04	7.28e-07	0.257	5.35e-04	8.60e-04	6.38e-04	7.39e-07	0.201	3.43e-07
rf	0.669	0.555	0.114	-0.009 [-0.032, 0.011]	no	8.44e-04	7.01e-04	7.12e-07	0.245	5.40e-04	7.55e-04	5.50e-04	5.70e-07	0.163	2.78e-08
extratrees	0.666	0.576	0.090	0.011 [-0.008, 0.026]	no	8.24e-04	6.83e-04	6.79e-07	0.238	5.17e-04	7.59e-04	5.53e-04	5.77e-07	0.161	-4.00e-20
mlp	0.638	0.602	0.036	0.037 [0.016, 0.053]	yes	7.99e-04	6.59e-04	6.38e-07	0.223	4.98e-04	7.90e-04	5.76e-04	6.24e-07	0.179	-7.68e-06
gp	0.604	0.572	0.032	0.008 [-0.011, 0.023]	no	8.28e-04	6.72e-04	6.85e-07	0.233	4.98e-04	8.26e-04	6.01e-04	6.82e-07	0.191	-4.37e-06

best-by-test-R2 = **svr** (0.678) · linear 0.625 · persistence 0.565 · H=900s · 2605.4s. te.* = OOS test losses; tr.* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

AI Sigma=30s | horizon=15m | metric=AI

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
linear	0.529	0.620	-0.091	0.055 [0.046, 0.063]	yes	7.81e-04	6.29e-04	6.09e-07	0.222	4.10e-04	9.02e-04	6.79e-04	8.13e-07	0.245	1.88e-15
ridge	0.529	0.620	-0.091	0.055 [0.046, 0.063]	yes	7.81e-04	6.29e-04	6.09e-07	0.222	4.10e-04	9.02e-04	6.79e-04	8.13e-07	0.245	-1.74e-16
lasso	0.528	0.616	-0.087	0.051 [0.042, 0.058]	yes	7.85e-04	6.33e-04	6.16e-07	0.223	4.14e-04	9.02e-04	6.79e-04	8.14e-07	0.244	1.49e-19
elasticnet	0.528	0.611	-0.083	0.046 [0.037, 0.054]	yes	7.89e-04	6.37e-04	6.23e-07	0.224	4.19e-04	9.03e-04	6.80e-04	8.15e-07	0.243	1.32e-19
huber	0.521	0.659	-0.138	0.094 [0.090, 0.098]	yes	7.39e-04	5.86e-04	5.47e-07	0.208	3.26e-04	9.09e-04	6.70e-04	8.26e-07	0.256	-1.01e-04
poly2	0.563	0.615	-0.051	0.050 [0.037, 0.062]	yes	7.86e-04	6.37e-04	6.17e-07	0.231	4.48e-04	8.68e-04	6.47e-04	7.53e-07	0.231	1.89e-14
poly3	0.567	0.591	-0.024	0.026 [0.011, 0.040]	yes	8.10e-04	6.60e-04	6.56e-07	0.252	4.84e-04	8.64e-04	6.44e-04	7.47e-07	0.264	-2.65e-13
knn	0.626	0.590	0.036	0.025 [0.010, 0.038]	yes	8.10e-04	6.65e-04	6.57e-07	0.227	4.95e-04	8.03e-04	5.86e-04	6.44e-07	0.183	-7.38e-06
svr ★	0.617	0.668	-0.051	0.103 [0.091, 0.114]	yes	7.30e-04	5.86e-04	5.32e-07	0.198	3.83e-04	8.13e-04	5.84e-04	6.61e-07	0.196	-9.27e-05
kernelridge	0.618	0.600	0.018	0.036 [0.018, 0.049]	yes	8.00e-04	6.55e-04	6.40e-07	0.256	4.91e-04	8.12e-04	5.94e-04	6.59e-07	0.248	4.60e-06
gbm	0.565	0.538	0.027	-0.027 [-0.037, -0.016]	no	8.60e-04	7.16e-04	7.40e-07	0.261	5.42e-04	8.66e-04	6.44e-04	7.50e-07	0.205	2.94e-07
rf	0.695	0.512	0.183	-0.053 [-0.083, -0.025]	no	8.84e-04	7.34e-04	7.82e-07	0.258	5.79e-04	7.25e-04	5.25e-04	5.26e-07	0.147	4.49e-07
extratrees	0.663	0.559	0.104	-0.006 [-0.029, 0.015]	no	8.41e-04	6.99e-04	7.06e-07	0.245	5.43e-04	7.62e-04	5.55e-04	5.81e-07	0.162	-3.93e-20
mlp	0.637	0.567	0.070	0.002 [-0.023, 0.025]	no	8.33e-04	6.90e-04	6.94e-07	0.237	5.37e-04	7.91e-04	5.78e-04	6.26e-07	0.179	2.86e-06
gp	0.603	0.567	0.036	0.002 [-0.011, 0.014]	no	8.33e-04	6.79e-04	6.94e-07	0.236	5.10e-04	8.27e-04	6.03e-04	6.85e-07	0.192	-3.34e-06

best-by-test-R2 = **svr** (0.668) · linear 0.620 · persistence 0.565 · H=900s · 2657.2s. te.* = OOS test losses; tr.* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

horizon=30m Sigma=30s · horizon=30m (H=1800s) — metrics: LE + AI

LE Sigma=30s | horizon=30m | metric=LE

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
linear	0.496	0.601	-0.105	0.067 [0.058, 0.074]	yes	1.11e-03	9.05e-04	1.24e-06	0.216	6.18e-04	1.28e-03	9.56e-04	1.64e-06	0.236	2.78e-15
ridge	0.496	0.601	-0.105	0.067 [0.058, 0.074]	yes	1.11e-03	9.05e-04	1.24e-06	0.216	6.18e-04	1.28e-03	9.56e-04	1.64e-06	0.236	-2.68e-17
lasso	0.496	0.597	-0.102	0.063 [0.055, 0.071]	yes	1.12e-03	9.09e-04	1.25e-06	0.217	6.21e-04	1.28e-03	9.57e-04	1.64e-06	0.236	1.43e-19
elasticnet	0.495	0.594	-0.099	0.060 [0.051, 0.067]	yes	1.12e-03	9.13e-04	1.26e-06	0.218	6.25e-04	1.28e-03	9.57e-04	1.64e-06	0.235	2.59e-19
huber ★	0.486	0.652	-0.166	0.118 [0.113, 0.123]	yes	1.04e-03	8.28e-04	1.08e-06	0.197	4.84e-04	1.29e-03	9.41e-04	1.67e-06	0.248	-1.58e-04
poly2	0.531	0.596	-0.066	0.062 [0.049, 0.075]	yes	1.12e-03	9.13e-04	1.25e-06	0.225	6.61e-04	1.24e-03	9.15e-04	1.53e-06	0.223	-5.88e-15
poly3	0.531	0.570	-0.038	0.036 [0.024, 0.047]	yes	1.16e-03	9.43e-04	1.34e-06	0.233	7.00e-04	1.24e-03	9.15e-04	1.53e-06	0.236	4.88e-13
knn	0.594	0.548	0.046	0.014 [-0.000, 0.025]	no	1.18e-03	9.77e-04	1.40e-06	0.229	7.36e-04	1.15e-03	8.32e-04	1.32e-06	0.183	-8.60e-06
svr	0.581	0.646	-0.065	0.112 [0.095, 0.127]	yes	1.05e-03	8.38e-04	1.10e-06	0.193	5.36e-04	1.17e-03	8.23e-04	1.36e-06	0.199	-1.60e-04
kernelridge	0.583	0.564	0.019	0.030 [0.015, 0.044]	yes	1.16e-03	9.61e-04	1.35e-06	0.248	7.30e-04	1.16e-03	8.45e-04	1.36e-06	0.230	8.27e-06
gbm	0.555	0.530	0.025	-0.004 [-0.014, 0.006]	no	1.21e-03	1.01e-03	1.46e-06	0.247	7.77e-04	1.20e-03	8.83e-04	1.45e-06	0.194	3.62e-07
rf	0.663	0.465	0.198	-0.069 [-0.100, -0.042]	no	1.29e-03	1.08e-03	1.66e-06	0.263	8.63e-04	1.05e-03	7.49e-04	1.10e-06	0.147	5.14e-07
extratrees	0.637	0.515	0.122	-0.019 [-0.041, -0.002]	no	1.23e-03	1.03e-03	1.51e-06	0.246	7.99e-04	1.09e-03	7.83e-04	1.18e-06	0.160	6.02e-21
mlp	0.603	0.557	0.046	0.023 [0.001, 0.041]	yes	1.17e-03	9.76e-04	1.37e-06	0.227	7.52e-04	1.14e-03	8.20e-04	1.29e-06	0.180	-9.27e-06
gp	0.572	0.551	0.021	0.017 [0.001, 0.031]	yes	1.18e-03	9.70e-04	1.39e-06	0.229	7.40e-04	1.18e-03	8.55e-04	1.39e-06	0.190	4.39e-07

best-by-test-R2 = **huber** (0.652) · linear 0.601 · persistence 0.534 · H=1800s · 2605.3s. te.* = OOS test losses; tr.* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

AI Sigma=30s | horizon=30m | metric=AI

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
linear	0.491	0.595	-0.104	0.061 [0.050, 0.070]	yes	1.12e-03	9.11e-04	1.26e-06	0.219	6.24e-04	1.29e-03	9.62e-04	1.66e-06	0.239	-1.66e-15
ridge	0.491	0.595	-0.104	0.061 [0.050, 0.070]	yes	1.12e-03	9.11e-04	1.26e-06	0.219	6.24e-04	1.29e-03	9.62e-04	1.66e-06	0.239	-3.02e-17
lasso	0.490	0.592	-0.101	0.058 [0.047, 0.067]	yes	1.13e-03	9.15e-04	1.27e-06	0.220	6.29e-04	1.29e-03	9.63e-04	1.66e-06	0.239	3.82e-19
elasticnet	0.490	0.588	-0.098	0.054 [0.044, 0.063]	yes	1.13e-03	9.19e-04	1.28e-06	0.221	6.33e-04	1.29e-03	9.63e-04	1.66e-06	0.238	2.47e-19
huber ★	0.481	0.645	-0.164	0.111 [0.106, 0.116]	yes	1.05e-03	8.36e-04	1.10e-06	0.201	4.88e-04	1.30e-03	9.47e-04	1.69e-06	0.252	-1.58e-04
poly2	0.525	0.581	-0.056	0.047 [0.031, 0.061]	yes	1.14e-03	9.30e-04	1.30e-06	0.230	6.74e-04	1.24e-03	9.22e-04	1.55e-06	0.226	6.41e-15
poly3	0.531	0.553	-0.023	0.019 [0.002, 0.034]	yes	1.18e-03	9.66e-04	1.39e-06	0.251	7.28e-04	1.24e-03	9.14e-04	1.53e-06	0.253	-8.16e-14
knn	0.592	0.546	0.046	0.012 [-0.006, 0.026]	no	1.19e-03	9.82e-04	1.41e-06	0.230	7.45e-04	1.15e-03	8.34e-04	1.33e-06	0.184	-1.33e-05
svr	0.579	0.642	-0.063	0.108 [0.097, 0.120]	yes	1.05e-03	8.49e-04	1.11e-06	0.196	5.63e-04	1.17e-03	8.27e-04	1.37e-06	0.200	-1.61e-04
kernelridge	0.583	0.557	0.026	0.023 [0.008, 0.037]	yes	1.17e-03	9.73e-04	1.37e-06	0.267	7.49e-04	1.16e-03	8.46e-04	1.36e-06	0.213	8.10e-06
gbm	0.552	0.528	0.024	-0.006 [-0.020, 0.005]	no	1.21e-03	1.02e-03	1.47e-06	0.250	7.88e-04	1.21e-03	8.88e-04	1.46e-06	0.196	2.62e-07
rf	0.660	0.436	0.225	-0.098 [-0.135, -0.066]	no	1.32e-03	1.11e-03	1.75e-06	0.270	8.98e-04	1.05e-03	7.54e-04	1.11e-06	0.148	5.24e-07
extratrees	0.634	0.494	0.140	-0.040 [-0.068, -0.014]	no	1.25e-03	1.05e-03	1.57e-06	0.254	8.38e-04	1.09e-03	7.87e-04	1.19e-06	0.161	4.89e-20
mlp	0.601	0.501	0.100	-0.033 [-0.066, -0.003]	no	1.24e-03	1.04e-03	1.55e-06	0.248	8.44e-04	1.14e-03	8.27e-04	1.30e-06	0.179	8.28e-06
gp	0.573	0.544	0.028	0.010 [-0.003, 0.023]	no	1.19e-03	9.80e-04	1.41e-06	0.232	7.53e-04	1.18e-03	8.56e-04	1.39e-06	0.191	-1.72e-08

best-by-test-R2 = **huber** (0.645) · linear 0.595 · persistence 0.534 · H=1800s · 2548.2s. te.* = OOS test losses; tr.* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

horizon=1h Sigma=30s · horizon=1h (H=3600s) — metrics: LE + AI

LE Sigma=30s | horizon=1h | metric=LE

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
linear	0.430	0.549	-0.119	0.064 [0.054, 0.073]	yes	1.63e-03	1.34e-03	2.67e-06	0.226	9.50e-04	1.85e-03	1.37e-03	3.43e-06	0.243	3.13e-16
ridge	0.430	0.549	-0.119	0.064 [0.054, 0.073]	yes	1.63e-03	1.34e-03	2.67e-06	0.226	9.50e-04	1.85e-03	1.37e-03	3.43e-06	0.243	3.54e-17
lasso	0.430	0.546	-0.117	0.062 [0.051, 0.070]	yes	1.64e-03	1.34e-03	2.68e-06	0.227	9.54e-04	1.85e-03	1.37e-03	3.43e-06	0.243	-2.85e-19
elasticnet	0.430	0.544	-0.114	0.059 [0.048, 0.067]	yes	1.64e-03	1.34e-03	2.70e-06	0.228	9.58e-04	1.85e-03	1.37e-03	3.43e-06	0.243	-9.71e-20

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
huber ★	0.417	0.621	-0.204	0.136 [0.129, 0.143]	yes	1.50e-03	1.19e-03	2.25e-06	0.200	7.18e-04	1.87e-03	1.34e-03	3.51e-06	0.259	-2.57e-04
poly2	0.463	0.529	-0.065	0.044 [0.029, 0.059]	yes	1.67e-03	1.37e-03	2.79e-06	0.237	1.01e-03	1.80e-03	1.32e-03	3.23e-06	0.231	1.52e-14
poly3	0.465	0.494	-0.029	0.009 [-0.006, 0.022]	no	1.73e-03	1.42e-03	3.00e-06	0.253	1.08e-03	1.79e-03	1.32e-03	3.22e-06	0.240	1.65e-12
knn	0.532	0.474	0.058	-0.011 [-0.028, 0.002]	no	1.76e-03	1.46e-03	3.11e-06	0.244	1.12e-03	1.68e-03	1.21e-03	2.81e-06	0.194	-1.98e-05
svr	0.511	0.616	-0.105	0.131 [0.119, 0.145]	yes	1.51e-03	1.21e-03	2.27e-06	0.196	7.90e-04	1.71e-03	1.19e-03	2.94e-06	0.216	-2.89e-04
kernelridge	0.520	0.499	0.021	0.014 [-0.003, 0.027]	no	1.72e-03	1.44e-03	2.97e-06	0.259	1.11e-03	1.70e-03	1.22e-03	2.89e-06	0.250	-8.76e-06
gbm	0.508	0.478	0.030	-0.007 [-0.021, 0.004]	no	1.76e-03	1.48e-03	3.09e-06	0.253	1.15e-03	1.72e-03	1.25e-03	2.96e-06	0.199	-1.07e-07
rf	0.586	0.400	0.186	-0.085 [-0.119, -0.057]	no	1.88e-03	1.58e-03	3.55e-06	0.272	1.26e-03	1.58e-03	1.13e-03	2.49e-06	0.169	2.18e-07
extratrees	0.583	0.426	0.157	-0.059 [-0.085, -0.039]	no	1.84e-03	1.55e-03	3.40e-06	0.265	1.22e-03	1.58e-03	1.13e-03	2.51e-06	0.167	1.05e-19
mlp	0.545	0.460	0.085	-0.025 [-0.055, 0.001]	no	1.79e-03	1.50e-03	3.20e-06	0.251	1.19e-03	1.65e-03	1.19e-03	2.73e-06	0.188	-6.62e-06
gp	0.508	0.464	0.043	-0.021 [-0.035, -0.008]	no	1.78e-03	1.47e-03	3.17e-06	0.249	1.14e-03	1.72e-03	1.24e-03	2.96e-06	0.202	-1.75e-05

best-by-test-R2 = **huber** (0.621) · linear 0.549 · persistence 0.485 · H=3600s · 2600.2s. te.* = OOS test losses; tr.* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

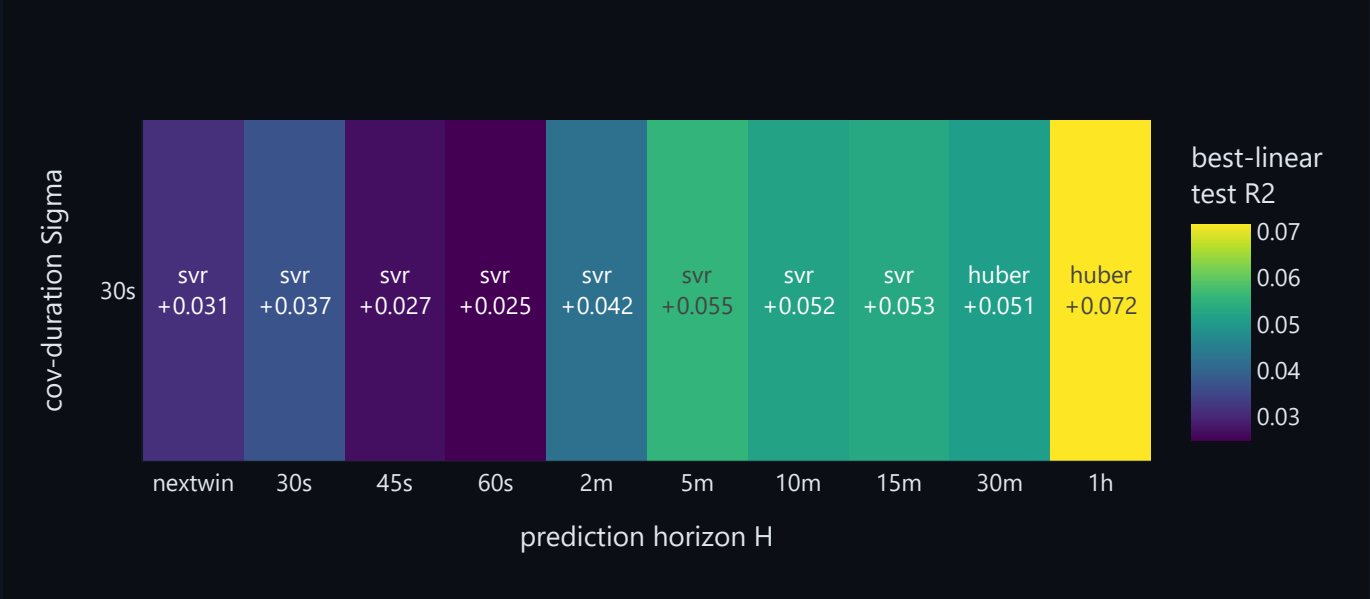
AI Sigma=30s | horizon=1h | metric=AI

learner	train R2	test R2	gap	dR2 [CI95]	sig	te.RMSE	te.MAE	te.MSE	te.QLIKE	te.bias	tr.RMSE	tr.MAE	tr.MSE	tr.QLIKE	tr.bias
linear	0.425	0.543	-0.118	0.058 [0.045, 0.069]	yes	1.64e-03	1.35e-03	2.71e-06	0.229	9.60e-04	1.86e-03	1.38e-03	3.46e-06	0.246	2.61e-16
ridge	0.425	0.543	-0.118	0.058 [0.045, 0.069]	yes	1.64e-03	1.35e-03	2.71e-06	0.229	9.60e-04	1.86e-03	1.38e-03	3.46e-06	0.246	3.53e-17
lasso	0.425	0.540	-0.115	0.055 [0.043, 0.066]	yes	1.65e-03	1.35e-03	2.72e-06	0.229	9.65e-04	1.86e-03	1.38e-03	3.46e-06	0.246	2.27e-19
elasticnet	0.425	0.537	-0.113	0.052 [0.040, 0.063]	yes	1.65e-03	1.35e-03	2.74e-06	0.230	9.69e-04	1.86e-03	1.38e-03	3.46e-06	0.246	-6.47e-21
huber ★	0.412	0.614	-0.202	0.129 [0.123, 0.136]	yes	1.51e-03	1.20e-03	2.28e-06	0.204	7.25e-04	1.88e-03	1.35e-03	3.54e-06	0.263	-2.57e-04
poly2	0.457	0.512	-0.055	0.027 [0.009, 0.045]	yes	1.70e-03	1.39e-03	2.89e-06	0.242	1.03e-03	1.81e-03	1.33e-03	3.26e-06	0.233	4.20e-14
poly3	0.466	0.478	-0.012	-0.007 [-0.027, 0.011]	no	1.76e-03	1.45e-03	3.09e-06	0.257	1.12e-03	1.79e-03	1.32e-03	3.21e-06	0.256	6.36e-13
knn	0.532	0.470	0.062	-0.015 [-0.037, 0.005]	no	1.77e-03	1.47e-03	3.14e-06	0.245	1.14e-03	1.68e-03	1.21e-03	2.81e-06	0.194	-2.57e-05
svr	0.511	0.598	-0.087	0.113 [0.100, 0.128]	yes	1.54e-03	1.24e-03	2.38e-06	0.203	8.30e-04	1.71e-03	1.19e-03	2.94e-06	0.215	-2.85e-04
kernelridge	0.521	0.486	0.035	0.001 [-0.019, 0.018]	no	1.74e-03	1.46e-03	3.04e-06	0.336	1.14e-03	1.70e-03	1.22e-03	2.88e-06	0.251	-1.08e-05
gbm	0.505	0.448	0.057	-0.037 [-0.057, -0.019]	no	1.81e-03	1.52e-03	3.27e-06	0.263	1.20e-03	1.73e-03	1.26e-03	2.98e-06	0.201	-1.26e-07
rf	0.586	0.340	0.246	-0.145 [-0.190, -0.107]	no	1.98e-03	1.65e-03	3.91e-06	0.284	1.34e-03	1.58e-03	1.13e-03	2.49e-06	0.170	1.54e-07
extratrees	0.583	0.392	0.191	-0.093 [-0.128, -0.060]	no	1.90e-03	1.60e-03	3.60e-06	0.274	1.29e-03	1.58e-03	1.13e-03	2.51e-06	0.168	4.21e-20
mlp	0.546	0.402	0.144	-0.083 [-0.123, -0.047]	no	1.88e-03	1.57e-03	3.54e-06	0.265	1.26e-03	1.65e-03	1.19e-03	2.73e-06	0.186	7.25e-06
gp	0.510	0.460	0.050	-0.025 [-0.042, -0.011]	no	1.79e-03	1.49e-03	3.20e-06	0.251	1.17e-03	1.72e-03	1.24e-03	2.95e-06	0.201	-1.66e-05

best-by-test-R2 = **huber** (0.614) · linear 0.543 · persistence 0.485 · H=3600s · 2489.9s. te.* = OOS test losses; tr.* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

Capacity - does a flexible learner beat linear? Per cell we compare the BEST learner's test R2 to the LINEAR learner's test R2. The heatmap shows the capacity gain (best - linear test R2) over (horizon x cov-duration), one panel per metric; the table lists the best learner per cell. A near-0 gain means the predictive map is essentially linear at that horizon; a positive gain means capacity helps.

Sigma=30sCapacity gain (best - linear test R2) [metric=LE]LE



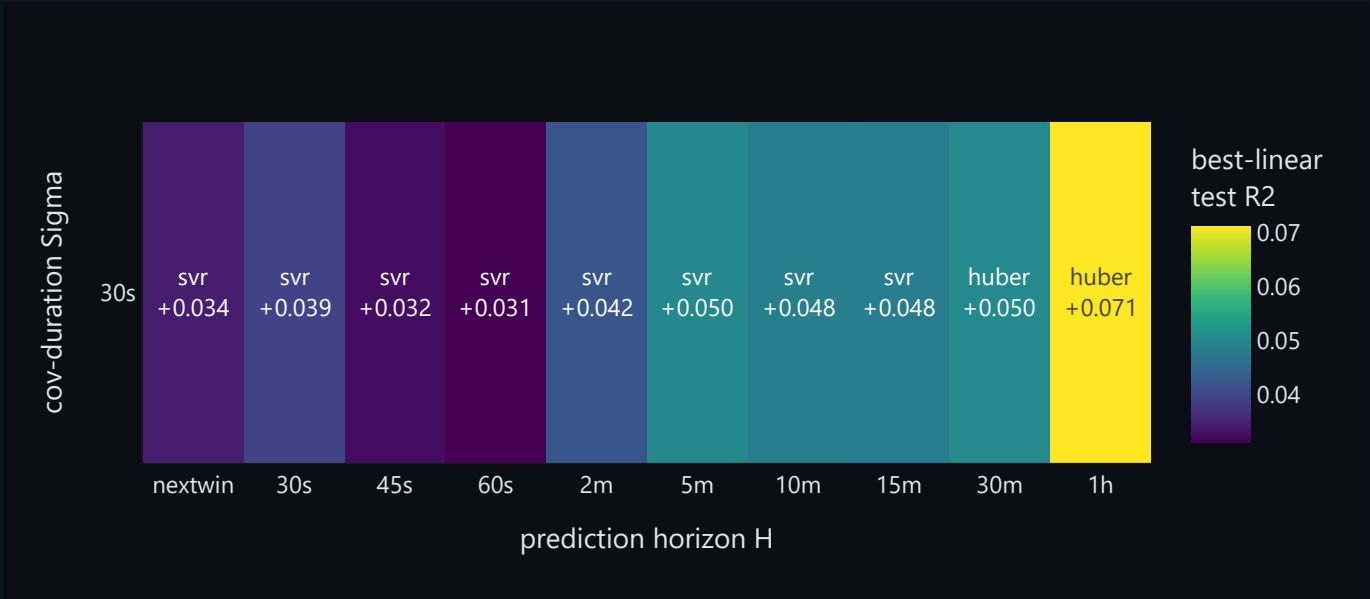
In plain terms How much a flexible learner beats plain linear at each cov-dur x horizon.

Math / formula gain = R2_OOS(best-by-test) - R2_OOS(linear); per (Sigma, H).

How to read it cell text = best learner + gain; bright/high = capacity helps, ~0 = linear is enough.

Insight metric LE; empty cells = not finished yet.

Sigma=30sCapacity gain (best - linear test R2) [metric=AI]AI



In plain terms How much a flexible learner beats plain linear at each cov-dur x horizon.

Math / formula gain = R2_OOS(best-by-test) - R2_OOS(linear); per (Sigma, H).

How to read it cell text = best learner + gain; bright/high = capacity helps, ~0 = linear is enough.

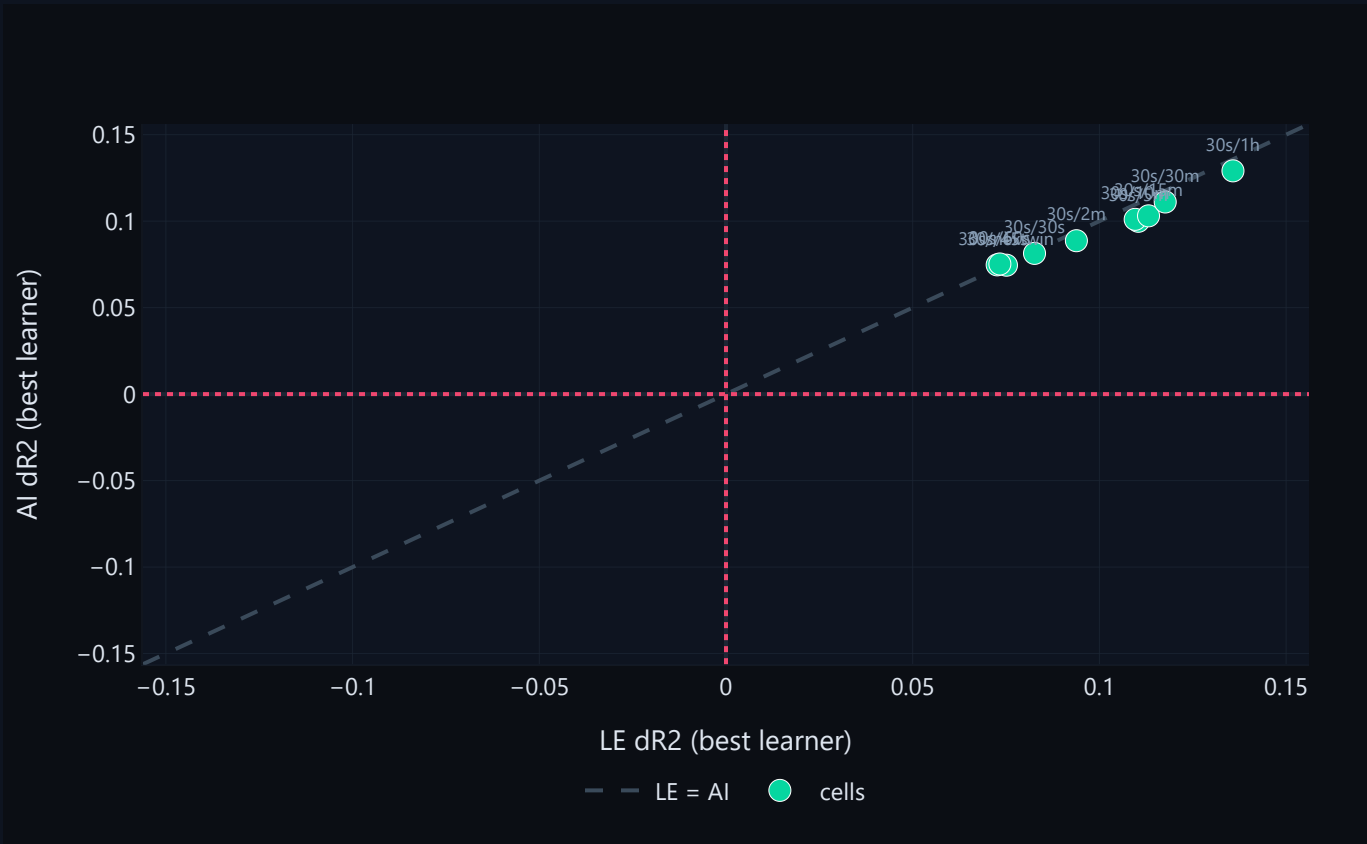
Insight metric AI; empty cells = not finished yet.

Best learner + capacity per cell (every finished cov-dur x horizon x metric).

Sigma	horizon	metric	best learner	best test R2	linear test R2	capacity gain	persistence R2	best overfit gap
30s	nextwin	LE	svr	0.502	0.471	0.031	0.427	-0.050
30s	nextwin	AI	svr	0.502	0.468	0.034	0.427	-0.052
30s	30s	LE	svr	0.505	0.468	0.037	0.423	-0.055
30s	30s	AI	svr	0.504	0.465	0.039	0.423	-0.056
30s	45s	LE	svr	0.495	0.468	0.027	0.422	-0.045
30s	45s	AI	svr	0.497	0.465	0.032	0.422	-0.049
30s	60s	LE	svr	0.558	0.533	0.025	0.485	-0.044
30s	60s	AI	svr	0.560	0.529	0.031	0.485	-0.048
30s	2m	LE	svr	0.629	0.587	0.042	0.536	-0.055
30s	2m	AI	svr	0.624	0.583	0.042	0.536	-0.053
30s	5m	LE	svr	0.682	0.627	0.055	0.572	-0.056
30s	5m	AI	svr	0.671	0.622	0.050	0.572	-0.048
30s	10m	LE	svr	0.682	0.630	0.052	0.573	-0.053
30s	10m	AI	svr	0.674	0.625	0.048	0.573	-0.046
30s	15m	LE	svr	0.678	0.625	0.053	0.565	-0.059
30s	15m	AI	svr	0.668	0.620	0.048	0.565	-0.051
30s	30m	LE	huber	0.652	0.601	0.051	0.534	-0.166
30s	30m	AI	huber	0.645	0.595	0.050	0.534	-0.164
30s	1h	LE	huber	0.621	0.549	0.072	0.485	-0.204
30s	1h	AI	huber	0.614	0.543	0.071	0.485	-0.202

LE vs AI - is the answer metric-robust? **AI** **LE** For each (cov-dur x horizon) with BOTH metrics finished, do LE and AI agree on (a) the best learner, and (b) the SIGN of the dial's dR2 over persistence? AI is the headline / LE the diagnostic (Step-6 convention). **metric_robust** = same dR2 sign AND both significant-or-both-not.

Sigma=30s LE vs AI dial edge (best-learner dR2)



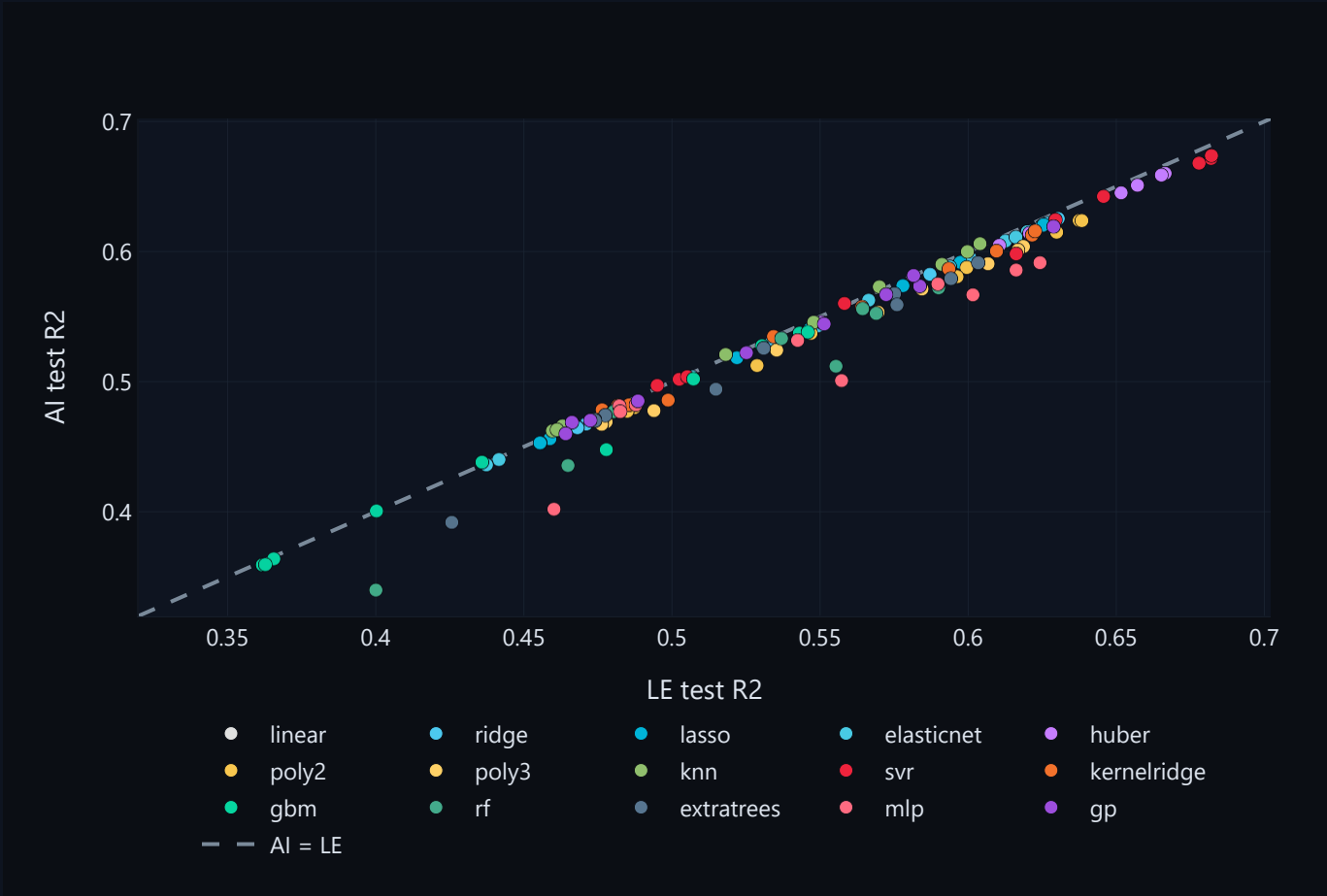
In plain terms Do the two SPD rulers (Log-Euclidean vs affine-invariant) agree on the dial's OOS edge?

Math / formula point = one (Sigma, H) cell; x = LE dR2, y = AI dR2; the diagonal is perfect agreement.

How to read it points on/near the diagonal and in the same quadrant = metric-robust; off-diagonal = fragile.

Insight clustering along y=x and in the +/- quadrant would confirm the AI dial reproduces the LE verdict.

Sigma=30s LE vs AI test-R2 per (cov-dur x horizon x learner) **AI** **LE**



In plain terms Every finished (cov-dur x horizon x learner) cell plotted as LE vs AI test R2.

Math / formula point = one (Sigma, H, model); x = LE test R2, y = AI test R2; dashed = AI=LE.

How to read it tight clustering on the diagonal = the SPD metric barely changes accuracy (metric-robust).

Insight color = learner; each spans all cov-dur x horizon cells where both metrics finished.

LE-vs-AI agreement per (cov-dur x horizon). same best learner? same dR2 sign? metric-robust?

Sigma	horizon	best LE	best AI	same best	dR2 LE	dR2 AI	same sign	metric-robust
30s	nextwin	svr	svr	True	0.075	0.075	True	True
30s	30s	svr	svr	True	0.083	0.081	True	True
30s	45s	svr	svr	True	0.073	0.075	True	True
30s	60s	svr	svr	True	0.073	0.075	True	True
30s	2m	svr	svr	True	0.094	0.089	True	True
30s	5m	svr	svr	True	0.110	0.100	True	True
30s	10m	svr	svr	True	0.110	0.101	True	True
30s	15m	svr	svr	True	0.113	0.103	True	True
30s	30m	huber	huber	True	0.118	0.111	True	True
30s	1h	huber	huber	True	0.136	0.129	True	True

Complete Step-7 performance picture. The full-matrix tab renders every learner; here we render the per-CELL summary + the per-YEAR breakdown of the best learner (train/test R2, dR2 over persistence, QLIKE, RMSE) for every finished (cov-dur x horizon x metric). Every row is labelled with cov-duration + horizon + metric. QLIKE is on variance; RMSE/MAE/bias in vol units.

Best-learner loss battery per cell (train R2 / test R2 / gap / dR2[CI] / sig / RMSE / MAE / QLIKE / bias / n_test).

Sigma	horizon	metric	best learner	train R2	test R2	gap	dR2 [CI95]	sig	RMSE	MAE	QLIKE	bias	n_test
30s	nextwin	LE	svr	0.453	0.502	-0.050	0.075 [0.070,0.080]	yes	2.24e-04	1.79e-04	0.777	5.10e-05	624446
30s	nextwin	AI	svr	0.450	0.502	-0.052	0.075 [0.070,0.079]	yes	2.24e-04	1.79e-04	0.747	5.04e-05	624446
30s	30s	LE	svr	0.451	0.505	-0.055	0.083 [0.078,0.087]	yes	2.18e-04	1.75e-04	0.968	4.53e-05	597064
30s	30s	AI	svr	0.448	0.504	-0.056	0.081 [0.077,0.086]	yes	2.18e-04	1.76e-04	0.921	4.49e-05	597064
30s	45s	LE	svr	0.450	0.495	-0.045	0.073 [0.066,0.079]	yes	2.20e-04	1.77e-04	0.739	5.01e-05	596378
30s	45s	AI	svr	0.448	0.497	-0.049	0.075 [0.070,0.079]	yes	2.19e-04	1.77e-04	0.706	4.93e-05	596378
30s	60s	LE	svr	0.514	0.558	-0.044	0.073 [0.067,0.080]	yes	2.58e-04	2.04e-04	0.365	8.90e-05	617379
30s	60s	AI	svr	0.512	0.560	-0.048	0.075 [0.070,0.081]	yes	2.58e-04	2.04e-04	0.360	8.92e-05	617379
30s	2m	LE	svr	0.575	0.629	-0.055	0.094 [0.087,0.101]	yes	3.08e-04	2.43e-04	0.309	1.20e-04	621119
30s	2m	AI	svr	0.571	0.624	-0.053	0.089 [0.082,0.096]	yes	3.10e-04	2.44e-04	0.302	1.24e-04	621119
30s	5m	LE	svr	0.626	0.682	-0.056	0.110 [0.102,0.118]	yes	4.26e-04	3.38e-04	0.226	1.98e-04	617436
30s	5m	AI	svr	0.624	0.671	-0.048	0.100 [0.091,0.108]	yes	4.33e-04	3.44e-04	0.230	2.06e-04	617436
30s	10m	LE	svr	0.629	0.682	-0.053	0.110 [0.100,0.119]	yes	5.88e-04	4.68e-04	0.200	2.88e-04	610631
30s	10m	AI	svr	0.627	0.674	-0.046	0.101 [0.090,0.111]	yes	5.96e-04	4.76e-04	0.204	3.02e-04	610631
30s	15m	LE	svr	0.619	0.678	-0.059	0.113 [0.103,0.123]	yes	7.18e-04	5.76e-04	0.194	3.67e-04	603869
30s	15m	AI	svr	0.617	0.668	-0.051	0.103 [0.091,0.114]	yes	7.30e-04	5.86e-04	0.198	3.83e-04	603869
30s	30m	LE	huber	0.486	0.652	-0.166	0.118 [0.113,0.123]	yes	1.04e-03	8.28e-04	0.197	4.84e-04	583830
30s	30m	AI	huber	0.481	0.645	-0.164	0.111 [0.106,0.116]	yes	1.05e-03	8.36e-04	0.201	4.88e-04	583830
30s	1h	LE	huber	0.417	0.621	-0.204	0.136 [0.129,0.143]	yes	1.50e-03	1.19e-03	0.200	7.18e-04	542965
30s	1h	AI	huber	0.412	0.614	-0.202	0.129 [0.123,0.136]	yes	1.51e-03	1.20e-03	0.204	7.25e-04	542965

Per-year OOS breakdown of the best learner (purged walk-forward test years). train R2 / test R2 / persistence R2 / dR2 / QLIKE / RMSE.

Sigma	horizon	metric	learner	test year	train R2	test R2	persistence R2	dR2	QLIKE	RMSE
30s	nextwin	LE	svr	2023	0.449	0.590	0.479	0.110	0.858	2.37e-04
30s	nextwin	LE	svr	2024	0.464	0.418	0.347	0.071	0.954	2.11e-04
30s	nextwin	LE	svr	2025	0.446	0.440	0.415	0.024	0.543	2.23e-04
30s	nextwin	AI	svr	2023	0.451	0.586	0.479	0.106	0.882	2.38e-04
30s	nextwin	AI	svr	2024	0.460	0.414	0.347	0.067	0.877	2.12e-04
30s	nextwin	AI	svr	2025	0.443	0.447	0.415	0.032	0.512	2.22e-04
30s	30s	LE	svr	2023	0.448	0.591	0.473	0.118	1.840	2.31e-04
30s	30s	LE	svr	2024	0.463	0.416	0.341	0.075	0.714	2.06e-04
30s	30s	LE	svr	2025	0.442	0.448	0.414	0.034	0.450	2.16e-04
30s	30s	AI	svr	2023	0.450	0.589	0.473	0.115	1.659	2.31e-04
30s	30s	AI	svr	2024	0.460	0.409	0.341	0.068	0.723	2.07e-04
30s	30s	AI	svr	2025	0.438	0.453	0.414	0.040	0.466	2.14e-04
30s	45s	LE	svr	2023	0.449	0.583	0.473	0.110	0.805	2.33e-04
30s	45s	LE	svr	2024	0.461	0.412	0.341	0.071	0.990	2.07e-04
30s	45s	LE	svr	2025	0.442	0.429	0.414	0.015	0.438	2.19e-04
30s	45s	AI	svr	2023	0.450	0.579	0.473	0.106	0.729	2.34e-04
30s	45s	AI	svr	2024	0.458	0.408	0.341	0.067	0.942	2.07e-04
30s	45s	AI	svr	2025	0.440	0.445	0.414	0.032	0.456	2.16e-04
30s	60s	LE	svr	2023	0.500	0.642	0.532	0.110	0.380	2.74e-04
30s	60s	LE	svr	2024	0.528	0.487	0.408	0.079	0.352	2.39e-04
30s	60s	LE	svr	2025	0.510	0.483	0.473	0.010	0.363	2.61e-04
30s	60s	AI	svr	2023	0.503	0.639	0.532	0.107	0.369	2.75e-04
30s	60s	AI	svr	2024	0.525	0.478	0.408	0.070	0.349	2.41e-04
30s	60s	AI	svr	2025	0.506	0.503	0.473	0.029	0.362	2.56e-04
30s	2m	LE	svr	2023	0.558	0.721	0.580	0.140	0.334	3.18e-04
30s	2m	LE	svr	2024	0.591	0.535	0.465	0.070	0.292	2.93e-04
30s	2m	LE	svr	2025	0.570	0.559	0.521	0.038	0.302	3.13e-04
30s	2m	AI	svr	2023	0.557	0.702	0.580	0.122	0.327	3.28e-04
30s	2m	AI	svr	2024	0.586	0.544	0.465	0.079	0.281	2.90e-04
30s	2m	AI	svr	2025	0.566	0.563	0.521	0.042	0.300	3.12e-04
30s	5m	LE	svr	2023	0.604	0.770	0.612	0.158	0.252	4.32e-04
30s	5m	LE	svr	2024	0.642	0.605	0.508	0.096	0.203	4.00e-04
30s	5m	LE	svr	2025	0.625	0.599	0.556	0.043	0.224	4.44e-04
30s	5m	AI	svr	2023	0.608	0.755	0.612	0.143	0.264	4.46e-04

Sigma	horizon	metric	learner	test year	train R2	test R2	persistence R2	dR2	QLIKE	RMSE
30s	5m	AI	svr	2024	0.638	0.583	0.508	0.075	0.206	4.10e-04
30s	5m	AI	svr	2025	0.620	0.604	0.556	0.048	0.221	4.41e-04
30s	10m	LE	svr	2023	0.599	0.781	0.609	0.172	0.220	5.84e-04
30s	10m	LE	svr	2024	0.649	0.589	0.514	0.075	0.180	5.62e-04
30s	10m	LE	svr	2025	0.628	0.591	0.559	0.032	0.200	6.18e-04
30s	10m	AI	svr	2023	0.603	0.767	0.609	0.158	0.230	6.03e-04
30s	10m	AI	svr	2024	0.647	0.560	0.514	0.046	0.187	5.82e-04
30s	10m	AI	svr	2025	0.623	0.610	0.559	0.051	0.197	6.03e-04
30s	15m	LE	svr	2023	0.585	0.774	0.599	0.175	0.214	7.21e-04
30s	15m	LE	svr	2024	0.639	0.578	0.509	0.069	0.175	6.90e-04
30s	15m	LE	svr	2025	0.619	0.599	0.553	0.046	0.191	7.43e-04
30s	15m	AI	svr	2023	0.588	0.760	0.599	0.161	0.224	7.42e-04
30s	15m	AI	svr	2024	0.636	0.549	0.509	0.040	0.180	7.14e-04
30s	15m	AI	svr	2025	0.614	0.610	0.553	0.057	0.189	7.32e-04
30s	30m	LE	huber	2023	0.440	0.672	0.563	0.109	0.257	1.21e-03
30s	30m	LE	huber	2024	0.497	0.631	0.484	0.147	0.167	8.96e-04
30s	30m	LE	huber	2025	0.498	0.634	0.526	0.108	0.167	9.86e-04
30s	30m	AI	huber	2023	0.438	0.660	0.563	0.097	0.263	1.23e-03
30s	30m	AI	huber	2024	0.491	0.625	0.484	0.141	0.170	9.03e-04
30s	30m	AI	huber	2025	0.492	0.636	0.526	0.110	0.169	9.84e-04
30s	1h	LE	huber	2023	0.354	0.636	0.507	0.128	0.267	1.76e-03
30s	1h	LE	huber	2024	0.429	0.608	0.439	0.169	0.167	1.27e-03
30s	1h	LE	huber	2025	0.434	0.605	0.484	0.122	0.166	1.42e-03
30s	1h	AI	huber	2023	0.354	0.622	0.507	0.115	0.274	1.80e-03
30s	1h	AI	huber	2024	0.423	0.603	0.439	0.163	0.170	1.28e-03
30s	1h	AI	huber	2025	0.429	0.610	0.484	0.126	0.167	1.41e-03

GridSearch best-params + wall-time per cell.

Sigma	horizon	metric	best learner	best_params	elapsed (s)
30s	nexttwin	LE	svr	{"C": 3.0, "gamma": 0.1}	2774.4
30s	nexttwin	AI	svr	{"C": 3.0, "gamma": 0.1}	2728.3
30s	30s	LE	svr	{"C": 3.0, "gamma": 0.1}	2676.5
30s	30s	AI	svr	{"C": 3.0, "gamma": 0.1}	2661.7
30s	45s	LE	svr	{"C": 3.0, "gamma": 0.1}	2681.8
30s	45s	AI	svr	{"C": 3.0, "gamma": 0.1}	2656.5
30s	60s	LE	svr	{"C": 3.0, "gamma": 0.1}	2688.0
30s	60s	AI	svr	{"C": 3.0, "gamma": 0.1}	2684.7
30s	2m	LE	svr	{"C": 3.0, "gamma": 0.1}	2735.9
30s	2m	AI	svr	{"C": 3.0, "gamma": 0.1}	2732.6
30s	5m	LE	svr	{"C": 3.0, "gamma": 0.1}	2655.5
30s	5m	AI	svr	{"C": 3.0, "gamma": 0.1}	2636.2
30s	10m	LE	svr	{"C": 3.0, "gamma": 0.1}	2654.0
30s	10m	AI	svr	{"C": 3.0, "gamma": 0.1}	2588.6
30s	15m	LE	svr	{"C": 3.0, "gamma": 0.1}	2605.4
30s	15m	AI	svr	{"C": 3.0, "gamma": 0.1}	2657.2
30s	30m	LE	huber	(no grid / defaults)	2605.3
30s	30m	AI	huber	(no grid / defaults)	2548.2
30s	1h	LE	huber	(no grid / defaults)	2600.2
30s	1h	AI	huber	(no grid / defaults)	2489.9

Step 7 (Track A) — horizon x cov-duration bake-off (PARTIAL render: 20 cell(s) finished)

Holding the 30/45/60 s masked covariance state fixed, the fixed Step-6 predictor `[rv, s_slow, s_fast, v]` is baked off with 15 learners against a realized-vol target at growing horizons, under LE and AI features. The headline object is **dR2(horizon)** - the dial's OOS edge over persistence - which is small/coincident at next-window and tests whether it grows (leading) with H. Current peaks: 30s/AI: peak dR2 0.129 @ 1h [sig]. **next-window is COINCIDENT; there is no next-day forecast here (deferred Track B).**

Holistic scoreboard — every finished cov-dur x horizon x metric, color-coded

GREEN = dial adds OOS value / robust / capacity helps / significant

AMBER = small / coincident / borderline / not significant

RED = dial hurts / dR2<0 / metric-fragile

cell (horizon · metric)	best learner	dR2 over persistence [CI95]	sig	best / linear test R2	capacity gain	QLIKE	LE-vs-AI
Sigma = 30s covariance window							
horizon=nextwin · LE	svr	0.075 [0.070,0.080]	sig	0.502 / 0.471	+0.031	0.777	robust
horizon=nextwin · AI	svr	0.075 [0.070,0.079]	sig	0.502 / 0.468	+0.034	0.747	robust
horizon=30s · LE	svr	0.083 [0.078,0.087]	sig	0.505 / 0.468	+0.037	0.968	robust
horizon=30s · AI	svr	0.081 [0.077,0.086]	sig	0.504 / 0.465	+0.039	0.921	robust
horizon=45s · LE	svr	0.073 [0.066,0.079]	sig	0.495 / 0.468	+0.027	0.739	robust
horizon=45s · AI	svr	0.075 [0.070,0.079]	sig	0.497 / 0.465	+0.032	0.706	robust
horizon=60s · LE	svr	0.073 [0.067,0.080]	sig	0.558 / 0.533	+0.025	0.365	robust
horizon=60s · AI	svr	0.075 [0.070,0.081]	sig	0.560 / 0.529	+0.031	0.360	robust
horizon=2m · LE	svr	0.094 [0.087,0.101]	sig	0.629 / 0.587	+0.042	0.309	robust
horizon=2m · AI	svr	0.089 [0.082,0.096]	sig	0.624 / 0.583	+0.042	0.302	robust
horizon=5m · LE	svr	0.110 [0.102,0.118]	sig	0.682 / 0.627	+0.055	0.226	robust
horizon=5m · AI	svr	0.100 [0.091,0.108]	sig	0.671 / 0.622	+0.050	0.230	robust
horizon=10m · LE	svr	0.110 [0.100,0.119]	sig	0.682 / 0.630	+0.052	0.200	robust
horizon=10m · AI	svr	0.101 [0.090,0.111]	sig	0.674 / 0.625	+0.048	0.204	robust
horizon=15m · LE	svr	0.113 [0.103,0.123]	sig	0.678 / 0.625	+0.053	0.194	robust
horizon=15m · AI	svr	0.103 [0.091,0.114]	sig	0.668 / 0.620	+0.048	0.198	robust
horizon=30m · LE	huber	0.118 [0.113,0.123]	sig	0.652 / 0.601	+0.051	0.197	robust
horizon=30m · AI	huber	0.111 [0.106,0.116]	sig	0.645 / 0.595	+0.050	0.201	robust
horizon=1h · LE	huber	0.136 [0.129,0.143]	sig	0.621 / 0.549	+0.072	0.200	robust
horizon=1h · AI	huber	0.129 [0.123,0.136]	sig	0.614 / 0.543	+0.071	0.204	robust

dR2 GREEN = positive AND 95% CI clears 0 (dial leads persistence here); AMBER = positive but not significant (coincident); RED = negative. Capacity gain = best-learner minus linear test R2. LE-vs-AI shows "pending" until the matching AI cell finishes (AI cells run after all LE cells).

What this establishes. Step 7 (Track A) traces the **horizon-of-predictability** curve for the covariance stress dial while holding the covariance state fixed. It converts Step-6's single next-window coincidence check into a full dR2(H) sweep and, for the first time, builds the **AI (affine-invariant) dial** that Steps 5/6 deferred - so the LE-vs-AI robustness claim is tested on a real predictor, not just a detector. The 15-learner bake-off separates "does the dial carry signal" from "does the answer need capacity" at each horizon.

Honest caveats / negatives.

- next-window is COINCIDENT.** At H=nextwin the dial co-moves with vol and persistence dominates; a small dR2 there is expected, not a failure.
- Longer horizons test LEADING, but overlapping targets inflate apparent skill** - the block-bootstrap CI over days (not the point estimate) is the honest significance test.
- No next-day forecast here.** Track B (daily HAR / QLIKE / Diebold-Mariano / MCS) is a separate deferred run; Track A stays intraday (nextwin..1h).
- The covariance state is unchanged** - no new Sigma window is built; this is a horizon sweep on the existing 30/45/60 s masked cloud.
- PARTIAL.** This render reflects only the cells finished so far (20); re-run the builder when the detached grid completes for the full scoreboard.